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FOR SOURCES SOUGHT PURPOSES ONLY

**Performance Work Statement (PWS)**

**for the**

**HUD Technology Transformation Next Generation**

**DATE: April 7, 2026**

**United States Department of Housing and Urban Development Office of Chief Information Officer**

**(US Department of HUD OCIO)**

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## C.1 SCOPE

Under the HUD Technology Transformation Next Generation Performance Work Statement (H2TNG PWS), the Contractor shall deliver comprehensive IT Support Services and Technology Transformation Next Generation (ITSS and 2TNG) solutions across HUD Headquarters (HQ), regional and field offices, data centers, and other designated locations. Support shall span the functional areas of the Office of Chief Information Officer (OCIO), including but not limited to:

- OCIO front office
- Assistance Chief Information Officer Business IT Resource Management (ACIO BIRM)
- Office of Chief Technology Officer (OCTO)
- ACIO Customer Relationship & Performance Management (ACIO CRPM)
- Office of Chief Information Security Office (OCISO)
- ACIO Office of Infrastructure & Operations (ACIO OIO)
- Office of the Artificial Intelligence Officer (OAIO)
- Office of Digital Business Officer (ODBO)

### C.1.1 Scope of Support

Individual TOs (TOs) may include support request for, but are not limited to:

Enterprise performance & risk management, performance management, project health assessments, risk planning, TechStat management, audit response development, communications, OCIO communications and channels, governance, annual strategic planning, customer care committee board, policy management, contract management, adjudication of contracts requesting onsite performance by contractors, acquisition management, acquisition coordination, asset requisition, COR support, administrative management, leadership and employee development and training support, telework/remote work support, reasonable accommodations, timekeeping, travel, space manage, supplies, workforce planning and talent management, financial management, budget execution, budget formulation and capital planning, micro-purchases, spend plan management, customer experience (CX), employee experience (EX), enterprise architecture, capabilities and standards, solution engineering, technical governance, technology strategy, technical solutions, analytics and reporting, cloud applications, custom development, data services, emerging technology, systems/software engineering, software technology demonstration and transitions, test and evaluation (T&E), independent verification & validation (IV&V), acquisition support, acquisition engagement and support, contract management support, budget formulation & capital planning, business need intake for pre-select phase, business need and case/select phase support, budget support, budget management and support, customer engagement, customer centric communication, customer relationship management, FITARA, FITARA request process facilitation, open government, audit and policy response facilitation, program & project management (P/PM), IT asset management, project lifecycle management, governance, risk & compliance, enterprise management framework (EMF), cybersecurity policies and procedures, FISMA CIO Quarterly/Annual Data Calls, plans of action and milestones (POA&Ms) tracking, security

assessments, cyber incident response team (CIRT), security operations center (SOC), threat intelligence, Information System Security Officer (ISSO) support, phishing exercises, security awareness and training, security architecture, data loss prevention, Zero Trust Architecture (ZTA), end-to-end lifecycle O&M support for program office systems, accounts & access, data center access management and support, DIAMS administration and support, DIAMS requests, asset management, hardware (H/W) and software (S/W) asset management, audit & reporting compliance, audit support, DHS/CIS BOD Reporting, configuration & release management, patch planning, existing device support, checkout inventory, Headquarters (H/Q) printer support, power outages, printer support, workstation moves, field support, enterprise application support, testing and deployment of new technologies, IT training, H/W, assistive technology (AT) & training resources, IT and operations & maintenance (O&M) support, IT and DME (development, modernization, and enhancement) support, IT and integration and deployment (I&D) support, end user support, network support, telecommunication support, updating outlook distribution lists, data center network security administration, data center network administration, email support, fiber cabling, IP address management, LAN administration, network operation pilots, regional and field office moves, remote access systems support, service desk ticket management, system/performance monitoring, system XID approvals, Tiger Team testing and troubleshooting, VoIP support, Video Teleconferencing (VTC) HQ support, WAN administration Wi-Fi administration, security management, antivirus management, Continuous Diagnostic Monitoring (CDM) administration, Enterprise Mobility Management (EMM), log monitoring and event management, security assessment, security configuration settings management (CSM), vulnerability scanning, system support, data center application support, data center asset and quality management and support, data center cloud management and support, data center database administration, data center mainframe administration, data center middleware administration, data center network administration, data center production control, data center security compliance, data center server administration, event support, IT facilities; and other solutions encompassing the entire spectrum of IT and IT requirements, to include software and hardware incidental to the solutions.

Contractors shall deliver scalable, secure, compliant, and customer-focused solutions under individual TOs that specify outcomes, service levels, and delivery locations.

### **C.1.2 IT Product Acquisition**

Individual orders may include the acquisition of software and hardware IT products necessary to implement the required solution or capability when such products aren't readily available through HUD's existing Blanket Purchase Agreements (BPAs) for hardware, software, and/or cloud services.

- The H2TNG multiple award vehicle is not intended to serve solely as a vehicle for purchasing IT products; however, products may be acquired when they are essential to delivering the specified solution or capability.
- After delivery, these products will be transitioned to HUD's Hardware, Software, and Cloud BPAs for ongoing lifecycle support and replacement.
- ITSS, and related IT products, may cover the full system lifecycle.

Contractors must be prepared to provide services and deliverables nationwide, including the United States (U.S.), the U.S. commonwealths, and unincorporated U.S. territories.

### **C.1.3 Functional Area Requirements**

The H2TNG PWS outlines general requirements, while specific requirements will be defined in individual TOs. Functional area requirements, described in Section 4.0 - Technical Function Areas, are not mutually exclusive; many requirements will span multiple areas to deliver a comprehensive, end-to-end lifecycle solution.

For this contract, end-to-end lifecycle refers to all phases of the product lifecycle, including:

- Development, Modernization, & Enhancement (DME)
- Integration & Deployment (I&D)
- Operations & Maintenance (O&M)
- Decommissioning and End-of-Life (EOL)

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## C.2 APPLICABLE DOCUMENTS

The Contractor shall comply with the documents listed below. Additional documents may be listed in individual TOs.

1. 44 U.S.C. § 3554, “Federal Information Security Modernization Act of 2014”
2. 44 U.S.C. 3551-3558, “Cybersecurity Improvements to Agency Information Systems”
3. 41 U.S.C. § 1321-1328, Federal Acquisition Supply Chain Security Act.
4. Federal Information Processing Standards (FIPS) Publication 140-3, “Security Requirements for Cryptographic Modules”, April 1, 2022
5. ISO/IEC 19790:2025, Information Security, cybersecurity and privacy protection – Security requirements for cryptographic modules, Edition 3, 2025.
6. 18 U.S.C. § 278g-3a, Internet of Things Cybersecurity Act.
7. 44 U.S.C. § 3607-3616, FEDRAMP Authorization Act
8. FIPS Pub 199. “Standards for Security Categorization of Federal Information and Information Systems,” February 2004
9. FIPS Pub 200, “Minimum Security Requirements for Federal Information and Information Systems,” March 2006
10. FIPS Pub 201-3, “Personal Identity Verification of Federal Employees and Contractors,” January 24, 2022
11. 10 U.S.C. § 2224, "Defense Information Assurance Program"
12. 42 U.S.C. § 2000d “Title VI of the Civil Rights Act of 1964”
13. Cyber Information Security Agency Binding Operational Directive 22-01 and its successors
14. HUDAR 2452.239-70 (Access to HUD systems) Part of HUD Procurement Handbook 2210.3, March 31, 2017
15. NIST SP 800-83 Rev 1, Guide to Malware Incident Prevention and Handling for Desktops and Laptops, July 2013
16. NIST SP 1800-25, Data Integrity: Identifying and Protecting Assets Against Ransomware and Other Destructive Events, December 2020
17. NIST SP 800-61 Rev 3, Computer Security Incident Handling Guide, April 3, 2025
18. NIST SP 800-53 Rev 5, Security and Privacy Controls for Information Systems Organizations, September 2020
19. OMB M-21-31, Improving the Federal Government’s Investigative and Remediation Capabilities Related to Cybersecurity Incidents
20. OMB Memorandum M-22-09, "Moving the U.S. Government Toward Zero Trust Cybersecurity Principles
21. OMB M-25-22, "Driving Efficient Acquisition of Artificial Intelligence in Government

22. OMB M-19-17, "Enabling Mission Delivery through Improved Identity, Credential, and Access Management
23. OMB Memorandum M-19-03, "Strengthening the Cybersecurity of Federal Agencies by Enhancing the High Value Asset Program,"
24. Executive Order 14306, June 6, 2025, "Sustaining Select Efforts to Strengthen the Nation's Cybersecurity and Amending Executive Order 13694 and Executive Order 14144,"
25. HUD CPO, AS-2315 Cybersecurity and Privacy Requirements (APRIL 2022)
26. HUD Handbook 2400.1 - Information Resources Management (IRM) Policies
27. HUD Handbook 2400.25 - Information Technology Security Policy
28. HUD Handbook 2400.25 - Vulnerability Disclosure Policy
29. HUD Handbook 3150.1 - Breach Notification Policy and Response Plan Rev 1.0
30. HUD Handbook 3250.1 - HUD Business Process Improvement
31. HUD Handbook 3251.1 - Secure Configuration Management Policy
32. HUD Handbook 3252.1 - Software Configuration Management Policy
33. HUD Handbook 3253.1 - Change Management Policy
34. HUD Handbook 3254.1 - Enterprise Patch Management Policy
35. HUD Handbook 3255.1 - Enterprise Architecture Policy
36. HUD Handbook 3256.1 - Information Technology Asset Management Policy
37. HUD Handbook 3260.1 - Enterprise Data Management (EDM) Policy
38. HUD Handbook 3262.1 - Software Acquisition Capability Maturity Model Policy
39. HUD Handbook 300.00.1 - Total Information Quality Management
40. HUD Handbook 3400.1 - HUD Policy for IT Management Framework
41. HUD Handbook 3410.1 - HUD Policy for Information Technology Governance
42. HUD Handbook 3417.1 - Web Application Policy
43. HUD Handbook 3425.1 - HUD Policy for Information Technology Strategic Planning
44. HUD Handbook 3430.1 - HUD Policy for Agile Methodology Rev. 1
45. HUD Handbook 3435.1 - HUD OCIO Risk Management Policy
46. HUD Handbook 3440.1 - HUD OCIO Techstat Policy
47. HUD Handbook 3500.1 - HUD Source Code Policy
48. HUD IT Capital Planning and Investment Control Process Guide, October 2020
49. HUD Handbook 2300.2 - Travel Handbook
50. HUD Handbook 1325.1 - HUD Privacy Handbook Rev. 1.0
51. HUD Handbook 1750.1 - Information and Administrative Security Program
52. HUD Handbook 1751.1 - HUD Controlled Information Program
53. HUD Handbook 2135.1 - Forms Management
54. HUD Handbook 2225.6 - HUD Records Disposition Schedules
55. HUD Handbook 2228.2 - HUD Records & Information Management Handbook
56. HUD Handbook 3230.2 - Physical Security Handbook for HUD Regional and Field Offices



57. 36 C.F.R. Part 1194 “Information and Communication Technology Standards and Guidelines,” January 18, 2017
58. Office of Management and Budget Circular A-130, “Managing Federal Information as a Strategic Resource,” July 28, 2016
59. 5 U.S.C. Section 552a, as amended, the Privacy Act of 1974
60. NIST SP 800-66 Rev. 2, “An Introductory Resource Guide for Implementing the Health Insurance Portability and Accountability Act (HIPAA) Security Rule,” February 2024
61. Sections 504 and 508 of the Rehabilitation Act (29 U.S.C. Section § 794d), as amended, January 18, 2017
62. Homeland Security Presidential Directive (12) (HSPD-12), August 27, 2004
63. National Institute of Standards and Technology (NIST) Special Publication (SP) 800-53 Rev 5, “Security and Privacy Controls for Federal Information Systems and Organizations,” September 23, 2020 (includes updates as of 12/10/2020)
64. Federal Travel Regulation (FTR). 2025 ([www.gsa.gov/federaltravelregulation](http://www.gsa.gov/federaltravelregulation))
65. Common Approach to Federal Enterprise Architecture and the Collaborative Planning Methodology (CPM), May 2012NIST SP 800-37 Rev 2, “Risk Management Framework for Information Systems and Organizations: A System Life Cycle Approach for Security and Privacy,” December 2018
66. OMB Memorandum M-24-15, Modernizing the Federal Risk and Authorization Management Program, July 25, 2024OMB Memorandum 05-24, “Implementation of Homeland Security Presidential Directive (HSPD) 12 – Policy for a Common Identification Standard for Federal Employees and Contractors,” August 5, 2005
67. OMB Memorandum M-19-17, “Enabling Mission Delivery Through Improved Identity, Credential, and Access Management,” May 21, 2019
68. GSA ICAM Solutions and Shared Services Roadmap ([GSA ICAM Solutions and Shared Services Roadmap - IDManagement](#))
69. OMB Memorandum, “Guidance for Homeland Security Presidential Directive (HSPD) 12 Implementation,” May 23, 2008
70. FICAM Architecture and Accompanying Playbooks, Version 3.3, June 30, 2023 ([FICAM Architecture - IDManagement](#))NIST SP 800-116 Rev 1, “Guidelines for the Use of Personal Identity Verification (PIV) Credentials in Facility Access,” June 2018
71. NIST SP 800-63-4, 800-63A-4, 800-63B-4, 800-63C-4, “Digital Identity Guidelines,” updated August 1,2025
72. NIST SP 800-157 Rev. 1, “Guidelines for Derived PIV Credentials,” November 2024
73. Trusted Internet Connections (TIC) 3.0 Core Guidance Documents, <https://www.cisa.gov/publication/tic-30-core-guidance-documents>
74. OMB Memorandum M-19-26, “Update to the Trusted Internet Connections (TIC) Initiative,” September 12, 2019
75. OMB Memorandum M-22-09, Federal Zero Trust Strategy, January 26, 2022
76. OMB Memorandum M-08-23, “Securing the Federal Government’s Domain Name System Infrastructure,” August 22, 2008
77. OMB Memorandum M-23-10, DOTGOV Online Trust in Government Act of 2020, February 8, 2023

78. Sections 524 and 525 of the Energy Independence and Security Act of 2007, (Public Law 110–140), December 19, 2007
79. Executive Order 13221, “Energy-Efficient Standby Power Devices,” August 2, 2001
80. Clinger-Cohen Act of 1996, 40 U.S.C. §11101, § 11331, and §11103
81. NIST SP 500-267B Revision 1, “USGv6 Profile,” November 2020
82. OMB Memorandum M-21-07, “Completing the Transition to Internet Protocol Version 6 (IPv6),” November 19, 2020
83. ANSI/EIA-310, Cabinets, Racks, Panels, and Associated Equipment, December 1<sup>st</sup>, 2005
84. ANSI/TIA-607-C, Telecommunications Bonding and Grounding (Earthing), November 2015
85. NFPA 70, National Electric Code (NEC), 2026
86. BICSI N3-2020, Bonding and Grounding, January 7, 2020
87. NFPA 17/17A, Dry & Wet Chemical Systems, 2024
88. NFPA 101, Life Safety Code, 2024
89. ASTM E814 & UL 1479, Standard Test Method for Fire Tests of Penetration Firestop Systems, May 2024
90. ASTM E1966 & UL 2079, Standard Test Method for Fire-Resistive Joint Systems, October 2024
91. NFPA 221, Standard for High Challenging Fire Walls, Fire Walls, and Fire Barrier Walls, 2024
92. NEC 406.3, Receptacle Rating and Type, 2023
93. NECA/NEMA 105-2024, Standard for Installing Metal Cable Tray Systems, 2024
94. ANSI/TIA-568-E-2020, Commercial Building Telecommunications Cabling Standards, 2020
95. ANSI/TIA-569, Telecommunication Pathways and Spaces, June 2022
96. ANSI/BICSI N2-3029, Installation of Telecommunications and ICT Cabling, December 2017
- 97. NEC Article 392, Cable Trays, P/O NFPA 70, 2026**

### **C.3 GENERAL REQUIREMENTS**

The Contractor shall provide and/or acquire ITSS and 2TNG solutions in support of the HUD OCIO relevant functional areas, together with any incidental hardware and software required under the individual Task Orders (TOs). Place of Performance

The place of performance will be specified in individual TOs. Work may occur at Government or non-Government sites within the United States, U.S. commonwealths, and unincorporated U.S. territories. Locations may include, but are not limited to, HUD HQ's, regional and field offices, data centers, and other designated sites as defined in each of the individual TOs.

#### **C.3.1 Travel**

Travel to HUD HQ's, regional and field offices, data centers, and/or other designated locations may be required under the H2TNG multiple award vehicle or individual TOs. Materials, Equipment and Locations

##### **C.3.1.1 Government-Furnished**

Government Furnished Property (GFP) – including Government Furnished Material (GFM), Government Furnished Information (GFI), and Government Furnished Equipment (GFE) may be provided and will be identified in individual TOs. Upon receipt, the Contractor shall conduct all necessary examinations, inspections, maintenance, and testing. The Contractor is responsible for reporting inspection results, maintenance actions, losses, and any damage to the Government through HUD's Software Asset Management (SAM) and/or Hardware Asset Management (HAM) modules, or other required asset management repositories specified in the individual TOs, as well as to the assigned Contracting Officers Representative (COR) and Program/Project Manager (P/PM).

HUD may provide HUD-specific software as required in individual TOs. The Contractor may use HUD-provided software development and test accounts, document repositories, and other resources necessary for development, storage, maintenance, and delivery of products. Contractors must comply with all HUD security policies and procedures for protecting sensitive data. Refer to Section 6.0 for detailed security requirements.

##### **C.3.1.2 Contractor-Acquired**

Upon Government approval, the Contractor shall acquire and/or provide any hardware or software required to fulfill each TO that is not supplied as GFP. The Contractor shall maintain software integrity in accordance with the producer's licensing agreement until the software is delivered to the Government or otherwise disposed of as directed by the Government. All items delivered must receive prior Government approval and comply with the requirements outlined in the H2TNG PWS. Refer to Section 6.0 for detailed security requirements.

### **C.3.1.3 Non-Developmental Items and Commercial Processes**

Non-Developmental Items (NDI), Commercial-Off-The-Shelf (COTS), and Government-Off-The-Shelf (GOTS) products shall be utilized to the maximum extent possible. The Contractor shall apply federal government – approved and commercially available – federal government approved industry best processes, standards, and technologies wherever feasible.

### **C.3.1.4 Connectivity**

HUD will provide connectivity to HUD-specific systems and network as required for task execution through HUD-approved remote access technology. This remote access will enable connectivity to HUD-specific software and development tools authorized for use on the HUD network. HUD may install equipment at the Contractor's site to ensure compliance with security requirements. The Contractor must meet all of HUD's security standards outlined in Section 2.0 and will bear the cost of providing connectivity to HUD data centers, HUD HQ's, regional, and field offices, and other designated locations as required and specified in individual TOs. Additional connectivity to HUD systems may be authorized as appropriate within individual TOs.

### **C.3.1.5 Government and Non-Government Facilities**

HUD HQ's, regional and field offices, data centers, and other designated locations may be made available for the performance of individual TOs. Contractor personnel may also perform at HUD-approved contractor facilities or remote locations if specified in the TO. Contractors may be required to establish operational support at these locations and must comply with all HUD and Federal Assessment and Authorization (A&A) requirements. Specific facilities will be identified in each TO, and the Contractor shall disclose facility details during the Request for Proposal (RFP) process for each TO. All facilities must be approved by HUD and comply with the security and privacy requirements outline in the H2TNG PWS Section 6.0 and the standards specified in Section 2.0. HUD facilities are located within the U.S., U.S. territories, and U.S. commonwealths.

### **C.3.1.6 Warranty**

Items acquired under individual TOs may require warranty protection. Any applicable commercial warranties shall be transferred to the Government. The type of warranty and extent of coverage will be determined for each TO.

### **C.3.1.7 Marking, Handling, Storage, Preservation, Packaging, Tracking & Shipping**

The Contractor shall establish and maintain procedures, in accordance with the H2TNG PWS Section 2.0 – Applicable Documents for handling, storage, preservation, packaging, marking, tracking, and shipping of all products. These procedures must ensure the protection of product quality and prevent damage, loss, deterioration, degradation, or substitution. The Contractor shall document these procedures and make them available for Government review and approval.

upon request by the CO, COR, P/PM, or designated representative. Documentation shall include process descriptions, quality control measures, and compliance verification records.

#### **C.3.1.8 Export Control**

The Contractor shall comply with all applicable laws and regulations governing export-controlled information and technology. The Contractor shall not use, distribute, transfer, or transmit any technology – including technology incorporated into products, software, or other information except in full compliance with these laws and regulations. Additionally, the Contractor is responsible for planning, obtaining, and maintaining all required export licenses necessary to fulfill the requirements of individual TOs.

#### **C.3.2 Safety and Environmental**

Safety and environmental requirements will be specified in individual TOs. The Contractor shall comply with the Office of Federal Sustainability Acquisition and Electronics Stewardship initiatives as outlined in each TO and in accordance with the policies referenced at <https://www.sustainability.gov/resources-eo-efo.html>.

#### **C.3.3 Enterprise and IT Framework**

##### **C.3.3.1 HUD Technical Reference Model**

For HUD-specific TOs, the Contractor shall support the HUD Enterprise Management Framework (EMF). In alignment with this framework, the Contractor shall comply with the US Department of HUD Office of Chief Information Officers Technical Reference Model (HUD OCIO TRM) and consider the HUD Enterprise Technology Strategic Plan. The HUD's OCIO TRM is a key component of the overall Enterprise Architecture (EA), providing a common vocabulary and structure for describing the IT used to develop, operate, and maintain enterprise applications. Additionally, the OCIO TRM – which includes the Standards Profile and Product List – serves as a technology roadmap and a tool for guiding and supporting OCIO initiatives. The Contractor is responsible for ensuring that all proposed solutions, designs, and implementations adhere to TRM standards and align with HUD's enterprise architecture principles. Compliance shall be documented and made available for Government review upon request by the CO, COR, IT P/PM, technical point of contact (TPOC), and or subject matter expert (SME).

##### **C.3.3.2 Federal Identity, Credential, And Access Management (FICAM)**

The Contractor shall ensure Commercial Off-The-Shelf (COTS) product(s), software configuration and customization, and/or new software are Personal Identity Verification (PIV) card-enabled by accepting Homeland Security Presidential Directive-12 (HSPD-12) credentials. This shall be implemented using HUD's Enterprise Technical Architecture (ETA), [https://www.ea.oit.va.gov/EAOIT/VA\\_EA/Enterprise\\_Technical\\_Architecture.asp](https://www.ea.oit.va.gov/EAOIT/VA_EA/Enterprise_Technical_Architecture.asp) and HUD Identity and Access Management (IAM) approved enterprise design and integration patterns. All Contractor delivered applications and systems must comply with HUD Identity, Credential, and

Access Management (ICAM) policies and guidelines, the HSPD-12 Program, and align with the Federal Identity, Credential, and Access Management (FICAM) Architecture v3.3 and accompanying playbooks.

Contractor-delivered applications and systems provide user authentication services compliant with the National Institute of Standards and Technology Special Publication (NIST SP) 800-63-3 – Digital Identity Guidelines. At a minimum, authentication must include Public Key Infrastructure (PKI)-based authentication supporting PIV cards and/or Common Access Card (CAC), as determined by the business needs.

All applications and systems must conform to the Identity and Access Management PIV requirements outlined in the Office of Management and Budget (OMB) Memoranda M-05-24, M-19-17, and NIST Federal Information Processing Standard (FIPS) 201-3. NIST SP 800-217 – Guidelines for Personal Identity Verification (PIV) Federation. Contractor-delivered applications and systems shall be listed on the FIPS 201-3 Approved Product List (APL). If not currently listed, the Contractor is responsible for completing the FIPS 201 Evaluation Program.

The Contractor shall ensure all Contractor delivered applications and systems support the following capabilities:

- Automated provisioning using HUD's enterprise provisioning service.
- Integration with HUD's authoritative identity source for provisioning identity attributes.
- Use of HUD defined unique identifiers (Secure Identifier [SEC ID] / Integrated Control Number [ICN]).
- Support for multiple authenticators per identity and authenticators at every Authenticator Assurance Level (AAL) appropriate for the solution.
- Identity proofing for each Identity Assurance Level (IAL) appropriate for the solution.
- Federation for each Federation Assurance Level (FAL) appropriate for the solution, if applicable.
- Two-factor authentication (2FA) using HUD Enterprise Design Patterns.
- A Security Assertion Markup Language (SAML) implementation for assertion-based authentication; additional assertion methods may be provided if compliant with NIST SP 800-63-3 – Digital Identity Guidelines.
- Authentication/account binding based on trusted Hypertext Transfer Protocol (HTTP) headers if trust-based authentication is used.
- Role Based Access Control (RBAC).
- Auditing and reporting capabilities.
- All FICAM efforts shall comply with all federal government standards, policies, and guidelines specified in Section 2.0 – Applicable Documents; and any new federal government standards, policies, guidelines, and executive orders created within the lifetime of this contract.

The required Assurance Levels for this contract effort are Identity Assurance Level (IAL) 3, Authenticator Assurance Level (AAL) 3, and Federation Assurance Level (FAL) 3.

### **C.3.3.3 Internet Protocol Version 6 (IPv6)**

The Contractor solution shall fully support Internet Protocol Version 6 (IPv6) in accordance with the Office of Management and Budget (OMB) memorandum issued on November 19, 2020 (OMB M-21-07). Compliance with IPv6 requirements shall follow the USGv6 Program (NIST USGv6 Program), NIST Special Publication (SP) 500-267B Revision 1-USGv6 Profile, and NIST SP 800-119 Guidelines for the Secure Deployment of IPv6.

IPv6 compliance shall be incorporated into all IT infrastructures, application designs as defined under O&M, development activities as defined under O&M, operational systems, sub-systems, and their integrations. All devices must support native IPv6 and dual-stack (IPv6 / IPv4) connectivity without requiring additional memory or resources from the Government, ensuring functionality in mixed environments. Furthermore:

- All public/external facing servers and services (e.g., web, email, DNS, ISP services, etc.) shall support native IPv6 and dual stack (IPv6 / IPv4) operations.
- All internal infrastructure and applications shall communicate using native IPv6 and dual stack (IPv6 / IPv4) configurations.

The Contractor shall provide an IPv6 Compliance Plan detailing how these requirements will be met, including timelines, implementation strategies, and risk mitigation measures for each TOs specifying IPv6 requirements. Additionally, the Contractor shall conduct IPv6 functionality and security testing and submit test results to the Government for review and approval prior to deployment of IT systems and services specified within the individual TOs.

### **C.3.3.4 Trusted Internet Connection (TIC)**

The Contractor solution shall comply with the requirements outlined in Office of Management and Budget Memorandum M-19-26, "Update to the Trusted Internet Connections (TIC) Initiative", HUD Directive 6513 "Secure External Connections", and the TIC 3.0 Core Guidance Documents, including all Volumes and TIC Use Cases published by the Cybersecurity & Infrastructure Security Agency (CISA). Any deviations from these requirements must receive prior approval from HUD's Office of the Chief Information Security Officer (OCISO) and the Assistant Chief Information Officer for the Office of Infrastructure and Operations (ACIO OIO) offices.

### **C.3.3.5 Standard Computer Configuration**

The Contractor's IT end-user solution developed for use on standard HUD computers shall be fully compatible with and supported on HUD's standard operating environment, which currently includes Windows 11 (64bit), Microsoft Edge (Chromium based), and Microsoft 365 Apps for Enterprise, as well as any future versions of Windows adopted by HUD during the contract period. Applications delivered to HUD for deployment on Windows 11 workstations shall be delivered as signed .msi packages with switches for silent and unattended installation. Updates shall be delivered as signed .msp files to enable seamless deployment using Microsoft Endpoint Configuration Manager (CM) HUD's current desktop application deployment tool.



All software code must be signed using a certificate from a HUD-trusted signing authority. The Contractor shall ensure and certify that the solution functions as intended on a standard HUD computer configured with non-admin, standard user rights and hardened according to the United States Government Configuration Baseline (USGCB) and Defense Information Systems Agency (DISA) Secure Technical Implementation Guide (STIG) for the applicable client operating system.

The Contractor shall provide a Compatibility and Deployment Certification Report to the Government for review and approval prior to production deployment for all TOs specifying the execution of this requirement. This report shall include evidence of successful installation, update processes, functionality testing under HUD's standard configuration, and compliance with all security and configuration requirements for both the current and future Windows versions adopted by HUD.

### **C.3.3.6 Enterprise Management Framework**

The HUD Enterprise Architecture (HEAR) Library provides a comprehensive enterprise-wide view of HUD IT systems, encompassing tools, reports, databases, dashboards, and analytics. HEAR enables OCIO to monitor system health and performance while delivering intelligent analysis and trending insights that support proactive enterprise system management.

By leveraging real-time information, OCIO can make strategic, operational, and investment decisions that enhance IT service delivery-improving performance, availability, user experience, and reliability across the enterprise.

The contractor shall comply with all HEAR requirements and ensure full integration and alignment with HUD's enterprise architecture standards. The Contractor must support and maintain HEAR capabilities to enable OCIO's ability to monitor, analyze, and manage enterprise systems effectively.

HEAR supports a unified enterprise service management model, including release management, configuration management, change management, and incident management, all aligned with industry- standard IT Infrastructure Library (ITIL) service management best practices. The Contractor shall adhere to these ITIL-aligned processes and any additional frameworks specified in individual TOs.

### **C.3.3.7 Authoritative Data Sources**

The HUD Enterprise Architecture Repository (HEAR) is a key component of the overall Enterprise Architecture (EA). IT establishes a common framework for data taxonomy that defines the data architecture used to develop, operate, and maintain enterprise applications.

The Contractor shall comply with the Department's Authoritative Data Source (ADS) requirement which mandates that HUD systems, services, and processes across the enterprise access HUD data solely through official HUD ADSs where applicable, see below.



Information Classes that compose each ADS are documented in HEAR under the Data & Information domain.

The Contractor shall ensure that all delivered applications and system solutions support the following:

- Full integration with HUD's ADS framework.
- Consistent use of official ADSs for all data access and exchange
- Alignment with HUD's enterprise data taxonomy and standards specified in Section 2.0-Applicable Documents.

#### **C.3.3.8 Social Security Number (SSN) Reduction**

The Contractor solution shall support the Social Security Number Fraud Prevention Act (SSN FPA) of 2017, which prohibits the inclusion of SSNs on any document sent by mail.

Additionally, the Contractor support shall align with Section 240 of the Consolidated Appropriations Act (CAA) 2018, enacted March 23, 2018, which mandates that HUD discontinue using SSNs as the Primary Identifier in all HUD information systems.

The Contractor shall ensure that:

- Any new IT solution eliminates the use of SSNs as the Primary Identifier and replaces them with the Integrated Control Number (ICN) for all individuals in the HUD information systems.
- All delivered applications and systems integrate with HUD's identity management solution to support identity traits, including the use of ICN as the Primary Identifier.

SSN's may only be used to identify an individual if required to obtain information from a system outside HUD's jurisdiction, and only when no alternative identifier is available.

#### **C.3.4 Development Methodologies**

The Contractor shall support a Service-Oriented Architecture (SOA), a flexible set of design principles applied during system development and integration phases, as specified at the TOs level.

Any SOA-based architecture shall be deployed on a secure, scalable, interoperable, and dynamic platform that provides end-to-end visibility and manageability – from application services through networking components – and supports use across multiple domains.

For HUD-specific TOs, the Contractor shall support HUD efforts in accordance with the following framework and processes:

Visual Innovation Platform (VIP) and Product Lifecycle Management (PLM):

- VIP enhances collaboration and decision-making through visual tools (e.g. digital concept boards, etc.).
- PLM serves as a central hub for data, connecting engineering, design, manufacturing, and supply chain teams.
- VIP and PLM work together to integrate workflows and drive digital transformation:
  - VIP connects to PLM, pulling real-time product data to create interactive visual workspaces.
  - Teams use VIP's visual boards to ideate, review, and make decisions, which then updates the central PLM system.
- This integration bridges strategic vision (VIP) with operational execution (PLM), accelerating innovation and information technology (IT) maturity.

The Contractor shall perform duties consistent with processes defined in the HUD Process Asset Library (PAL), a centralized digital repository that stores and shares process-related resources (templates, guidelines, policies, checklists, lessons learned, and metrics).

PAL ensures:

- Standardized operations and compliance with HUD policies.
- Reuse of existing assets to reduce redundancy.
- Capturing organizational knowledge and facilitating continuous improvement.
- Clear role and responsibilities for HUD and Contractor staff.

The integration of PAL, PLM, and VIP creates a unified system that:

- Streamlines development and improves quality.
- Links design and capability to HUD's mission and vision.
- Breaks down silos, automates workflows, and ensures smooth data flows across all levels.

For HUD-specific TOs, the Contractor shall follow the HUD PAL, PLM, and VIP processes to ensure compliance with HUD policies, regulations, and quality standards. The Contractor shall use an incremental development methodology such as Agile (e.g. Scrum, Kanban, etc.), unless otherwise specified at the TOs level.

### **C.3.5 Integrated Product Teams**

The Contractor shall, when required by individual TOs, serve as a member of or provide Subject Matter Expertise (SME) to Integrated Product Teams (IPTs), Working IPTs (WIPTs), and/ or Integrated Business Teams (IBTs) within HUD. The Contractor's specific role(s), responsibilities, and level of participation shall be defined in each TOs. The Contractor shall actively participate in all assigned team activities and provide deliverables, recommendations, and technical input as requested by HUD.

IPTs, WIPTs, and IBTs are cross-functional teams that collaborate to develop strategies and approaches to achieve defined objectives. The Contractor shall support HUD in establishing and executing critical elements across all phases of the program and project lifecycle, including but not limited to:

- Acquisition Strategy.
- Requirements Development and Management.
- Use Case Development.
- Risk Management.
- Architecture Design.
- Performance Engineering.
- Capacity Planning.
- System and Software Development.
- Test and Evaluation.
- Sustainment Operations.

The Contractor shall ensure compliance with all HUD policies, standards, and applicable regulations while performing these duties. The Contractor shall provide timely, accurate, and complete documentation and reporting as required by HUD and specified in individual TOs.

### **C.3.6 Quality Assurance**

All Contractors and Subcontractors performing work under individually TOs shall possess the ISO 9001 Quality Management System (QMS) Certification at time of TO award and maintain this certification throughout the TO period of performance if specified within the TO.

Contractors shall comply with ISO 9001:2015 QMS requirements for obtaining and maintaining the ISO 9001 QMS Certification and shall submit a copy of the ISO 9001 QMS Certification with all Request for Proposal (RFP) responses for individual TOs awarded under this contract.

In addition, all Contractors and Subcontractors performing work on individual TOs shall possess the Capability Maturity Model Integration (CMMI) Maturity Level 5 – Optimizing with Capability Level 3 – Defined certification at time of TOs award and maintain this certification throughout the period of performance if specified within the TO. Contractors shall comply with CMMI requirements for obtaining and maintaining certification and shall submit a copy of their the CMMI Maturity Level 5 – Optimizing with Capability Level 3 – Defined certification with all Request for Proposal (RFP) responses for any individual TOs that specify it.

Certain Contractor and Subcontractor personnel may also be required to hold Information Technology Infrastructure Library (ITIL) 4 Certifications at one or more of the following levels: Level 1 – Foundation, Level 2 – Intermediate/Specialist (i.e. ITIL Managing Professional (MP), ITIL Strategic Leader (SL), and/or Practice Manager), Level 3 - ITIL Expert, and Level 4 – ITIL Master. Certification requirements for Contractor and Subcontractor staff will be identified in TOs labor categories, “key personnel” sections, or within individual TOs awarded under this contract. Contractor and Subcontractors shall submit copies of required ITIL 4 Certifications to the Contracting Officers Representative (COR). All Key Personnel shall submit resumes to the COR for review, ensuring that all names and PII are removed prior to submission. All Key

Personnel submitted as part of the proposal to the multiple-award vehicle or subsequent TOs, if awarded, must be on the contract a minimum of 4 months from the time they receive GFE.

### **C.3.7 Transition and Orientation Support**

The Contractor shall provide transition and orientation services, including the development of a Phase-In/Phase-Out Transition Plan, to ensure continuity of services as specified in the individual TOs. Transition and orientation support may include transferring support services to Government personnel or to another Contractor, as required, to maintain uninterrupted operations.

### **C.3.8 Government Inspection and Oversight**

The Contractor shall cooperate with authorized Government offices in matters related to facilities access, audits, security incident notification, and hosting location requirements. The Contractor and any Subcontractors shall:

- a. Provide the CO, Contracting Officers Representative (COR), TOs Program/Project Manager (P/PM), Technical Points of Contact (TPOC), and other designated representatives by the CO, COR, P/PM, and/or TPOC, full and unrestricted physical and remote/logical access to their facilities, installations, operational documentation, databases, and personnel associated with hosting services under this contract and individual TOs. This access shall be granted as necessary to conduct audits, inspections, device scanning using Government-approved tools, investigations, or other reviews to ensure compliance with contractual requirements for IT and information security and to safeguard against threats and hazards to the integrity, availability, and confidentiality of agency information under the Contractor's or Subcontractor's control.
- b. The Contractor shall fully cooperate with all audits, inspections, investigations, or other reviews conducted by or on behalf of the CO or other authorized Government offices. Full cooperation includes, but is not limited to, timely disclosure of requested data, information, and records; making Contractor employees available for interviews; and providing prompt access to facilities, systems, data, and personnel as reasonably required to complete the review. All cooperation shall be provided at no additional cost to the Government. The Contractor shall preserve all data, records, logs, and other evidence, reasonably necessary, to conduct a thorough investigation of any computer security incident, as defined in NIST SP 800-61, including incidents that pose an actual or potential threat to the integrity, availability, or confidentiality of agency information or the functionality of information systems operated under this contract. The Contractor shall promptly notify the CO, COR, and P/PM of any computer security or privacy incident in addition to any other notification requirements specified by law or this contract. Federal agency timeframes for reporting incidents to the United State Computer Emergency Readiness Team (US-CERT), as referenced in NIST SP 800-61 Rev. 3, shall serve as a guideline for determining timeliness.
- c. Upon request, the Contractor shall provide Government data, information, or records under its control to the CO, COR, P/PM, TPOC, or other authorized Government offices, including the Office of Inspector General (OIG) in support of audits, inspections,

investigations, reviews or litigation involving HUD. Requests shall specify a compliance deadline of not less than ten (10) days, and unless otherwise stated in this contract or individual TOs, production shall be at no additional cost to the Government.

- d. The Contractor shall include the substance of this section in all subcontracts where Subcontractor employees will have access to agency information or information systems operated under this contract and individual TOs. Furthermore, the Contractor shall ensure that all hosting services and storage of agency data, information, and records under this contract occur within the United States of America.

FOR SOURCES SOUGHT PURPOSES ONLY

## **C.4 TECHNICAL FUNCTIONAL AREAS**

Individual TOs may span multiple functional and sub-functional areas. Additional details regarding these areas are provided in subsequent sections to offer greater insight into the complexity and uniqueness of certain potential TOs and contract requirements outlined in this PWS. These requirements are not mutually exclusive and may apply across multiple functional and sub-functional areas.

If the Contractor's efforts under this contract are of such a nature that they could create a potential organizational conflict of interest (OCI), as defined in Federal Acquisition Regulation (FAR) Subpart 9.5, Contractor personnel may be required to sign a Non-Disclosure Agreement (NDA).

The following is applicable to all functional areas within the H2TNG multiple-award vehicle. The Contractor shall perform all relevant services associated with a functional area in alignment with applicable Federal and HUD policies, governance processes, and HUD approved industry frameworks, as specified in individual TOs.

### **C.4.1 Program Management, Strategy, Enterprise Architecture and Planning Support**

The Contractor shall provide comprehensive Program and Project Management (P/PM) services, including monitoring, analysis, strategic planning, and Enterprise Architecture (EA) support at the enterprise, portfolio, program, and individual project levels.

#### **C.4.1.1 Strategy and Planning**

The Contractor shall provide services that facilitate strategic decision-making regarding the organization's current and future IT structure and program integration. Services shall include, but are not limited to:

- Systematic assessment and redesign of key technologies, business processes, activity-based costing, and organizational structures.
- Streamlining processes, aligning organizational structures to the way work is conducted and deploying proven supporting technologies where appropriate.
- Developing recommendations that contribute to an overarching IT strategy aligned with business goals, objectives, and initiatives, and that leverage innovation to identify new opportunities for success.
- Designing measurable objectives and metrics aligned to the overall IT strategy and operations.

#### **C.4.1.2 Compliance**

The contractor shall ensure that all recommendations, analyses, supporting documentation and plans shall comply with Federal legislation and be consistent with Federal policies, standards,

and guidelines, including but not limited to: the Government Performance and Results Act (GPRA), Clinger-Cohen Act, Federal Activities Inventory Reform (FAIR) Act, and Government Paperwork Elimination Act (GPEA). Where applicable, recommendations shall align with agency policies and enterprise governance standards; and Section 2.0 of this PWS. The Contractor shall monitor and incorporate updates to relevant laws, regulations, and guidance throughout the period of performance, as directed at the Task Order (TO) level.

#### **C.4.1.3 Inherently Governmental Functions**

All Contractors and Subcontractors supporting Strategy and Planning efforts shall ensure their staff comply with FAR Subpart 7.5 - Inherently Governmental Functions. The Contractor shall not perform inherently governmental functions and shall immediately report any suspected violations to the contract CO, COR, and P/PM.

#### **C.4.2 Standards, Policy, Procedure and Process Development; and Implementation Support**

The Contractor shall support the development and/or evaluation of new standards, policy directives, operating procedures, processes, and related assessments, including impact analyses associated with implementation.

##### **C.4.2.1 Inherently Governmental Functions**

All Contractors and Subcontractors supporting these efforts shall ensure compliance with FAR Subpart 7.5 - Inherently Governmental Functions and shall immediately report suspected violations to the CO, COR, and P/PM.

#### **C.4.3 Requirements Development and Analysis Support**

The Contractor shall provide requirements development and analysis support as required by individual TOs. For iterative and agile methodologies, requirements may be identified and/or evolve at any phase of the lifecycle; therefore, requirements definition shall be structured to meet the incremental delivery needs.

Requirements support may include, but is not limited to:

- Enterprise analysis.
- Business and application architecture.
- Business process reengineering.
- Feasibility studies.
- Requirements planning and management.
- Requirements gathering/elicitation.
- Use Case development.
- Agile requirements methods.
- Requirements analysis.

- Change management.
- Peer reviews.
- Solution assessment and validation.
- Business process modeling and workflow management.

#### **C.4.3.1 Requirements Packages**

The Contractor shall provide requirements package support that may include, but is not limited to:

- Assistance in developing Statements of Objectives (SOO), Statements of Work (SOW), Performance Work Statements (PWS), Performance Specifications, Rough Orders of Magnitudes (ROM), Independent Government Cost Estimates (IGCE), Quality Assurance Surveillance Plans (QASP), and associated acquisition documentation.
- Technical advice and assistance regarding proposal evaluation (excluding inherently governmental decisions).
- Market research, evaluation, and recommendations of technical alternatives.

#### **C.4.3.2 Inherently Governmental Functions**

All Contractors and Subcontractors supporting Requirements Package efforts shall ensure compliance with FAR Subpart 7.5 - Inherently Governmental Functions and shall immediately report suspected violations to the CO, COR, and Government P/PM.

#### **C.4.4 Technology Refresh and Configuration Reviews**

The Contractor shall perform technology refresh and configuration reviews for any IT system, structure, or process to realize innovations that provide for business or technical improvements. Technology refresh efforts shall:

- Support formal procedures to manage business and technical innovations.
- Ensure new innovations are reviewed and adopted as required.
- Provide domain-specific recommendations commensurate with unique organizational requirements.

#### **C.4.5 Studies and Analyses**

The Contractor shall perform studies and analyses which may include, but are not limited, logistics/supportability, engineering, financial, operational, business processes, mobile applications, analytics, modernization of existing systems and applications, and interoperability/information sharing across Federal agencies and public/private systems.



#### **C.4.5.1 Technology Evaluation**

The Contractor shall conduct periodic engineering studies and analyses to evaluate the viability of potential solutions, alternatives to technical issues and challenges, and emerging products or technologies. This includes evaluating unproven technology applications, identifying potential risks, and developing pre-production or COTS-based prototypes, as required.

#### **C.4.6 Program Management Support**

The Contractor shall provide program management support to accomplish the administrative, managerial, logistical, integration and financial aspects (Program, Planning, Budget and Execution (PPBE)) specified in individual TOs.

##### **C.4.6.1 Key Personnel**

The Contractor shall identify an individual as the primary point of contact for all program-related issues, concerns, and status. This individual shall be designated Key Personnel for the contract and individual TOs.

##### **C.4.6.2 Government Decision Authority**

Decision-making resides with the Government. The Contractor may be required to provide subject matter expertise to coach, mentor, and/or consult with Government Program Managers to plan and execute processes prescribed by industry and government best practices consistent with organizational policies and procedures, as specified in individual TOs.

##### **C.4.6.3 Project/Product Management Functions**

The Contractor shall support project/product management functions and reporting, including, but not limited to:

- Project planning.
- Schedule management.
- Financial management.
- Earned Value Management (EVM).
- Quality management.
- Resource management.
- Requirements management.
- Communications management.
- Project change management.
- Organizational change management.
- Risk management.
- Performance management.
- Knowledge management (KM).

#### **C.4.7 Product Data**

The Contractor shall review, develop and/or submit Product Data identified in individual TOs as deliverables. Product Data may define configuration items, associated processes and procedures, and other items throughout the applicable lifecycle.

##### **C.4.7.1 Types of Product Data**

Product Data includes, but is not limited to: engineering drawings, form/fit/function requirements, design documentation, specifications, software configuration documentation, software code, interface control documentation, management/project plans, reports and analyses, VIP and PLM artifacts, quality assurance provisions, and/or commercial item descriptions.

##### **C.4.7.2 Tools and Methods**

The Contractor may be required to apply Computer-Aided Design (CAD) (i.e. AutoCAD, SolidWorks, and Catia), and Computer Automated Engineering (CAE) (i.e. prototyping and simulation) methods/systems to support concurrent design integration with manufacturing and logistics considerations.

#### **C.4.8 IT Services Management Support**

The Contractor shall recommend and support the development of IT service management (ITSM) plans, practices, infrastructures and systems utilizing industry best practices (e.g. Information Technology Infrastructure Library (ITIL)) to minimize negative impact on the IT enterprise.

IT services management includes, but is not limited to:

- Change management.
- Release management.
- Configuration Management (CM).
- Incident management.
- Problem management.
- Service desk management.
- Network and application monitoring management.
- Availability management.
- Capacity management.
- Event management.
- Data & storage management.
- Service Level Agreements (SLA).

#### **C.4.9 Development Toolkits**

The Contractor shall recommend and support the utilization of development toolkits (e.g., Atlassian Jira/Confluence). A full list of HUD's development toolkit will be provided at the individual TOs level when required for the performance of the effort. The Contractor shall ensure proper licensing, access controls, and configuration consistent with agency security policies and governance standards.

#### **C.4.10 Systems/Software Engineering**

The Contractor shall provide engineering expertise to analyze system concept, system design and interoperability, and shall recommend optimization strategies. The Contractor shall review and analyze development, production, system design, monitoring, and support proposals. The Contractor shall review and analyze development, production, system design, monitoring, and support proposals. Activities shall include, but are not limited to:

- Conducting trade-off and best technical approach analyses, including IGCE's and cost benefit analysis (e.g., Return on Investment (ROI)).
- Performing analysis of alternatives and engineering studies.
- Developing System Engineering Plans (SEPs), design plans, and technical reports as specified in individual TOs.

The Contractor shall provide systems/software engineering support across any or all phases of the system/software lifecycle including but not limited to:

- Acquisition Strategy.
- Requirements Development and Management.
- Use Case Development.
- Risk Management.
- Architecture Design.
- Performance Engineering.
- Capacity Planning.
- System/Software Development.
- Test and Evaluation.
- Sustainment Operations.

Requirements development associated with iterative or agile methodologies may occur at any phase of the lifecycle. Therefore, requirements definitions shall be structured to meet the incremental delivery needs.

During the lifecycle process, software engineering support shall include, but is not limited to:

- Software system reliability assessments.
- Participation in governance boards, WIPTs, and IPTs.

- Ensuring dependencies, interoperability, availability, reliability, maintainability, and performance of systems within Government-provided guidelines.

#### **C.4.10.1 Business Process Modeling (BPM) and Business Process Reengineering (BPR)**

The Contractor shall provide BPM and BPR support, including developing activity and process models for requirement analysis and identification of improvement opportunities. Specific tasks, deliverables, modeling standards, tools, and performance expectations for BPM/BPR activities will be defined in individual Task Orders (TOs).

#### **C.4.10.2 Design and Development**

The Contractor shall provide services across all lifecycle phases, including planning, programming, requirements analysis, design, coding, unit testing, system integration testing, implementation, monitoring design, configuration, implementation, maintenance, and updates of systems, applications, and services. This includes, but is not limited to:

- Information processing systems.
- Payroll and financial management systems.
- Decision support systems.
- Workflow management systems.

The Contractor shall ensure alignment with HUD strategic plans, initiatives, and goals. Specific tasks, deliverables, performance standards, and technical requirements associated with this lifecycle support will be further defined in individual Task Orders (TOs).

#### **C.4.10.3 Architecture Development**

The Contractor shall support the enhancement of Enterprise Architectures (EA), design patterns and associated Technical Reference Models (TRM), as well as the development of Business Line Architectures (BLA) and Solution Architectures (SA). Support activities may include, but is not limited to:

- Developing processes and data models derived from HUD Strategic Plans.
- Designing common infrastructure components and shared services to support systems development and integration.
- Supporting the retirement of legacy systems and building scalable, extensible systems based on reusable services.
- Conducting architecture and infrastructure audits/assessments and gap analysis.
- Supporting governance processes for architecture strategies.

Specific architectural tasks, required artifacts, framework alignment, modeling tools, and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.10.4 IT Service Management (ITSM) Implementation**

The Contractor shall implement ITSM plans, practices, infrastructures, and systems utilizing industry best practices (e.g. ITIL). The goal of these activities is to minimize negative impact on the IT enterprise. ITSM activities include:

- Change management.
- Release management.
- Configuration Management (CM).
- Incident and problem management.
- Service desk management.
- Network and application monitoring management.
- Availability and capacity management.
- Event and data management.
- Storage management.
- SLA management.

Specific tasks required ITSM processes, performance standards, documentation requirements, and service-level expectations will be further defined in individual Task Orders (TOs).

#### **C.4.10.5 Enterprise Application and Services**

The Contractor shall perform requirements analysis, system analysis, development, and implementation support for core business and mission support applications. Activities may include but are limited to:

- Process re-engineering and adaptation of IT solutions.
- Advanced collaboration capabilities.
- Workflow and business process modeling/translation.
- System modeling and simulation.
- Software development.
- Executive dashboards and enterprise search and discovery.
- Project management and scheduling tools.
- Multi-media support for training and marketing.

Specific tasks, application environments, technical requirements, deliverables, toolsets, and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.10.6 Cloud Computing**

The Contractor shall create and implement cloud computing solutions and support existing cloud computing environments. Solutions shall include:

- Utilize approved SaaS offerings listed in the Digital HUD Marketplace.
- Be developed and sustained in the HUD Enterprise Cloud (HEC) and/or HUD Platform One (HPO) environments.
- Incorporate modern cloud-native approaches (e.g. serverless, containerization).
- Prioritize PaaS over IaaS where possible.
- Be cost-optimized, secure, resilient, and aligned with HEC/HPO capabilities.

HUD will furnish HEC and/or HPO capacity as GFE. The Contractor shall not be responsible for acquiring cloud capacity or hosting infrastructure. Specific cloud computing tasks, deliverables, and performance expectations will be defined at the individual Task Orders (TOs).

### **C.4.10.7 Web and Mobile Application Design and Development**

The Contractor shall provide full lifecycle services for evaluation, planning, requirements analysis, design, coding, unit testing, system integration testing, implementation, deployment, maintenance, and updates of web and mobile applications. This support includes:

- Web-based application or web-enabling a current or existing systems.
- Mobile applications, including distribution and ongoing support.

The Contractor shall ensure that all web and mobile solutions comply with HUD standards, usability requirements, security policies, and enterprise governance processes. Specific tasks, platform requirements, toolsets, security standards, deliverables, and performance expectations for web and mobile application development will be defined in individual Task Orders (TOs).

### **C.4.10.8 Human-Computer Interaction**

The Contractor shall provide services related to analysis, design, evaluation, implementation, and testing of user interfaces for optimized performance, user experience, and Section 508 compliance. Computer systems include but are not limited to interactive and/or natural language processing.

### **C.4.10.9 System/Software Integration**

The Contractor shall provide support planning, updating architecture models, interoperability specifications and analysis, system interface specifications, service definitions, and segmented architecture for the transition, integration, and implementation of IT systems. Specific tasks, deliverables, accessibility standards, testing requirements, and performance expectations will be defined in individual Task Orders (TOs).

### **C.4.10.10 Modeling and Simulation**

The Contractor shall provide the personnel, equipment, tools, and facilities necessary to model, simulate, and analyze IT services, systems, networks and infrastructure. Modeling and simulation activities may include, but are not limited to:

- Performance prediction based on variables (e.g. network latency, end-user device performance, and system-component upgrades).
- Prototype implementations.
- Mathematical modeling.

Specific modeling requirements, simulation tools, deliverables, technical standards, and performance expectations will be defined in individual Task Orders (TOs).

### **C.4.10.11 Informatics Services**

The Contractor shall provide informatics services, including:

- Evaluation, analysis, and recommendation of technology improvements and insertions.
- Development of KM strategies, policies, procedures, and best practices.
- Analysis of lifecycle cost, schedule, tradeoffs, interoperability, performance impacts, suitability, and other salient characteristics.
- Design, develop, implement, and maintenance of information management structures, systems, and applications.
- Analyze new requirements and existing systems to determine and identify separable functions that are common across systems and potentially can be accommodated by COTS software, or alternatively by custom developed software.
- Software engineering assessments on business process support systems to provide evaluation of modifiability, portability, reusability, performance, and other quality areas.
- Develop architecture guidelines to include system/software architecture, software development processes, management indicators/quality metrics, requirements specifications and documentation standards.
- Data architecture repository and data architecture services.

### **C.4.10.12 Engineering and Technical Documentation**

The Contractor shall prepare and/or update engineering, user, technical documentation, reports, and manuals for existing or newly developed projects, software applications, or systems. Documentation shall be accurate, complete, clear, and maintained in accordance with HUD standards, enterprise governance processes, and applicable Federal requirements. Specific documentation requirements, templates, formats, repositories, deliverables, reviews, and performance expectations will be defined in individual Task Orders (TOs).

### **C.4.10.13 Current System and Data Migration**

The Contractor shall support migration of current systems and data to a common architecture within the HUD Network. Migration activities ensure continuity of operations, data integrity, compatibility with the target environment, and alignment with HUD enterprise architecture and governance standards. Activities may include but not limited to:

- Software engineering and data management.
- Database architecture support for:
  - Computer resource management.
  - Technical documentation analysis.
  - Participation in technical reviews.
  - Evaluation of test plans.
  - System and integration testing.
  - Applicability studies and analysis of common software.
  - Decommissioning of current and legacy systems.

Specific migration tasks, data conversion requirements, validation procedures, tools, deliverables, and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.10.14 Development Toolkit Support**

The Contractor shall provide acquisition, installation, administration, and maintenance of development toolkits (e.g., Atlassian Jira/Confluence) as required by individual TOs.

#### **C.4.10.15 Site Reliability Engineering (SRE) Support**

The contractor shall provide expertise in Site Reliability Engineering (SRE) principles, including:

- Conducting analysis, drafting recommendations and developing implementation plans for improved reliability and resiliency.
- Developing and tracking of error budgets.
- Developing and implementing automation to reduce operational toil.
- Developing performance data-driven analysis and assessment of IT service performance trends, incidents, and recurring issues.
- Supporting major incident response, problem management, and root cause analysis (RCA).

#### **C.4.11 Software Technology Demonstration and Transition**

The Contractors shall provide demonstrations, assessments, and transition support for advanced emerging software technologies. This functional area includes evaluating existing and emerging software products against the requirements of current system development and support efforts, demonstrating specific technologies within the context of supported systems, and transitioning effective technology solutions into operational use.

Current technology areas of focus for HUD include, but are not limited to: software architectures, databases, web-based applications, mobile applications, enterprise solutions, wireless technologies, and security solutions.



This capability is a critical component of HUD's mission to enhance and advance its software engineering practices and overall IT modernization objectives. Specific demonstration tasks, evaluation criteria, prototype requirements, transition deliverables, and performance expectations will be defined in individual Task Orders (TOs).

### **C.4.12 Test & Evaluation (T&E)**

The Contractor shall provide Test & Evaluation (T&E) support throughout all phases of the systems and software development lifecycle to ensure quality, performance, compliance, and readiness for operational deployment. This includes, but is not limited to:

- Preparing comprehensive test plans and procedures.
- Design detailed tests cases.
- Conducting and witnessing tests.
- Providing technical support during test execution.
- Coordinating test activities in compliance with applicable regulations and standards.
- Analyzing, evaluating, and documenting test results.

Additionally, the Contractor shall participate in technical analyses, code reviews, and other reviews as required by individual TOs.

### **C.4.13 Independent Verification and Validation (IV&V)**

The Contractor may be required to provide an independent review of products developed by other entities for quality, security, performance, and compliance with required standards. This includes reviewing, evaluating, validating and verifying processes, procedures, and methodologies used in developing, testing, maintaining, and securing third-party systems and software. Specific IV&V tasks, documentation reviews, evaluation criteria, independent requirements, deliverables, and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.13.1 Independence & Governance**

The Contractor shall establish and maintain independent governance mechanisms that ensure IV&V activities are performed objectively and without organizational, financial, or technical conflict of interest with system development personnel. Independence shall be maintained throughout all phases of system development, testing, deployment, and operations

- The Contractor shall provide direct reporting of IV&V findings, risks, and recommendations to Government-designated oversight authorities.
- IV&V governance shall be integrated into enterprise Agile and DevSecOps delivery models by maintaining independence while supporting continuous planning, backlog refinement, and release decision processes.

Specific independence requirements, reporting expectations, governance procedures, and oversight mechanisms will be defined in individual Task Orders (TOs).

#### **C.4.13.2 Requirements Verification**

The Contractor shall verify and validate all functional, non-functional, security, and regulatory requirements to ensure completeness, correctness, traceability, and testability. For the verification activities, the contractor shall confirm that requirements are properly documented, feasible, aligned with agency enterprise goals, and are suitable for implementation.

- Identify and document requirement gaps, ambiguities, and conflicts prior to design and implementation.
- Perform continuous requirements verification within Agile and DevSecOps environments, ensuring user stories, epics, and acceptance criteria remain traceable to approved enterprise and statutory requirements across iterative releases.

#### **C.4.13.3 Design & Architecture Validation**

The Contractor shall validate system architecture and detailed designs to confirm alignment with approved requirements, security and privacy principles, scalability, resiliency, and maintainability objectives. Activities shall include, but are not limited to:

- Identify architectural risks, design deficiencies, and non-conformities prior to implementation.
- Support continuous architecture validation in Agile and DevSecOps pipelines, including incremental design changes, infrastructure-as-code, and evolving trust boundaries across frequent releases.

#### **C.4.13.4 Implementation Verification**

The Contractor shall independently verify software implementations, including source code, configurations, and infrastructure components, to confirm adherence to approved designs, secure coding standards, and configuration baselines and agency enterprise governance requirements. Activities shall include, but are not limited to:

- Identifying and documenting defects, vulnerabilities, and deviations with severity and remediation guidance.
- Integrating implementation verification into automated DevSecOps pipelines, leveraging continuous code scanning, configuration validation, and dependency analysis without impeding delivery velocity.

Specific implementation verification tasks, deliverables, and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.13.5 Test & Evidence Validation**

The Contractor shall validate the adequacy, coverage, and accuracy of all test plans, test cases, and test results, including functional, integration, performance, and security testing. Activities shall include, but are not limited to:

- Confirming that evidence provided is complete, reproducible, traceable, and sufficient to demonstrate requirement satisfaction.
- Validating continuously generated test evidence within Agile and DevSecOps environments, ensuring automated and manual test results support incremental releases and continuous authorization decisions.

Specific test-evidence validation tasks, traceability standards, tools, documentation requirements, deliverables, and performance requirements will be defined in individual Task Orders (TOs).

#### **C.4.13.6 Operational Readiness Validation**

The Contractor shall validate operational readiness by assessing monitoring, logging, incident response, backup and recovery, continuity procedures, and sustainment processes. Activities shall include, but are not limited to:

- Determining system readiness for deployment and ongoing operations under real-world conditions.
- Performing continuous operational readiness validation within DevSecOps delivery models, ensuring each release increment is supportable, observable, and resilient in production environments.

Specific operational readiness validation tools, acceptance criteria, documentation requirements, deliverables, and performance requirements will be defined in individual Task Orders (TOs).

#### **C.4.13.7 Security, Privacy, and Compliance Validation**

The Contractor shall independently validate the implementation and effectiveness of security, privacy, and compliance controls, confirm authorization boundary accuracy, and shall assess residual risk associated with deployment, sustainment, and continuous operation of the system. Activities shall include, but are not limited to:

- Supporting continuous security and compliance validation within Agile and DevSecOps pipelines, enabling ongoing risk assessment and authorization through automated, evidence-driven controls.

Specific validation tasks, evidence requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.13.8 Risk & Issue Management**

The Contractor shall identify, assess, prioritize, document, and track technical, security, and operational risks and issues, including likelihood and impact. Activities shall include, but are not limited to:

- Providing objective, independent recommendations for risk mitigation, remediation, risk reduction or formal risk acceptance by the Government.
- Integrating risk and issue management into Agile and DevSecOps ceremonies and tooling, providing near-real-time visibility into emerging risks across iterative development and release cycles.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.13.9 Reporting & Decision Support**

The Contractor shall provide clear, timely, and decision-oriented reports that translate IV&V findings into mission, operational, technical and risk impacts. Activities shall include, but are not limited to:

- Supporting Government milestone decisions, including release readiness, deployment, authorization, and sustainment determinations.
- Providing reporting compatible with Agile and DevSecOps delivery cadence, enabling informed release, sprint, and continuous authorization decisions through concise, evidence-based dashboards and summaries.

#### **C.4.14 Enterprise Network**

The Contractor shall provide comprehensive systems and network administration, infrastructure support, and data, voice, and video services to meet the organization requirements. Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.14.1 Systems/Network Administration**

The Contractor shall provide end-to-end support for the establishment, operation, administration, maintenance, migration, monitoring, analysis, and retirement of information systems, storage systems, network systems and security systems in locations within the U.S., U.S. territories and U.S. commonwealths.

This support includes systems supporting end-to-end Fault, Configuration, Administration, Performance, and Security (FCAPS) aspects of network management for IT equipment currently in use or under consideration for procurement by HUD.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.14.2 Network and Telecommunications Infrastructures**

The Contractor shall design, deliver, operate, monitor, maintain, transition, and decommission network and telecommunications solutions, including turn-key communications systems.

Services shall include, but are not limited to:

- Network planning and infrastructure design.
- Engineering, installations, configuration, testing, and maintenance of network and telecommunications components.
- Support for voice, data, and video networks, including converged networks and cloud-based solutions.
- Provision and integration of telecommunication services such as Voice-as-a-Service (VaaS) and integration with carrier-provided solutions.

##### **C.4.14.2.1 Data Communications Systems**

The Contractor shall provide services for secure and on-secure data communications systems, including but not limited to:

- Network management equipment.
- Asynchronous Transfer Mode (ATM) and Internet Protocol (IP) equipment.
- Switching systems, Private Branch Exchange (PBX) systems, and Call Center as a Service (CCaaS).
- Computer telephony interfaces, Channel Service Units (CSU)/Digital Service Units (DSU), wireless systems, encryption tools, interfaces, and signal conversion and interface equipment.

This includes all systems residing on the customer premises, beyond the carrier demarcation point. Specific data communication tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.14.2.2 Voice Systems**

The Contractor shall design, deliver, operate, monitor, maintain, transition, and decommission secure and non-secure voice systems supporting agency enterprise communications requirements, including but not limited to:

- Legacy Private Branch Exchange (PBX) systems.
- Hybrid Voice over Internet Protocol-Time Division Multiplex (VoIP-TDM) systems.
- VoIP Systems.
- Automatic Call Distribution (ACD) systems.

- Intelligent Call Routing Systems.
- Call center and Interactive Voice Response (IVR) Systems.

The contractor shall also support leased voice solutions from telecommunications carriers. Specific voice system tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

### **C.4.14.2.3 Video Systems**

The Contractor shall provide services for secure and non-secure video systems, including but not limited to:

- Closed Circuit Television (CCTV), Cable TV (CATV), Video Teleconference (VTC) and desktop Local Area Network (LAN) VTC systems.
- Web-based collaboration tools, technical and training support.

Components may include, but are not limited to cameras, recorders, multipoint bridges, Integrated Services Digital Network (ISDN) and dial up systems, amplifiers, microphones, compression equipment, equalizers, remote controls, special optical enhanced equipment, and video interface equipment.

Video distribution may be wireless or via fiber optics, coaxial cable, or twisted pair copper cable. Specific video system tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

### **C.4.14.2.4 Local Area Network (LAN)/Wide Area Network (WAN) Systems**

The Contractor shall design, deliver, operate, monitor, maintain, transition, and decommission secure and non-secure Local Area Network (LAN) and Wide Area Network (WAN) systems, including but not limited to:

- Inside and outside cable plant, including structured cabling, fiber-optic.
- Wireless LAN and WAN components including access points, wireless.
- Network devices and transport technologies including routers, Ethernet switches, multiplexers (e.g. Synchronous optical networking (SONET), Dense Wavelength Division Multiplexing (DWDM)), network test equipment, and network management systems.

### **C.4.14.2.5 Software Defined Networks**

The Contractor shall provide services related to Software-Defined Networking (SDN) solutions, including design, implementation, maintenance and transition. Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### C.4.14.2.6 Other Transmission Systems

The Contractor shall design, deliver, maintain and transition secure and non-secure transmission systems, including but not limited to:

- Single and multi-mode fiber optics.
- Fiber optic multiplexing equipment.
- Wireless, Radio Frequency (RF), satellite communication systems.
- Fiber-to-copper and copper-based links.
- Repeaters, switching- protection mechanisms, and encryption solutions.

#### C.4.15 Enterprise Management Framework

The Contractor shall provide services in support of executing the EMF. These services shall include, but not limited to:

- Design, develop, and maintain robust data pipelines using modern ETL/ELT (Extract, Transform, Load / Extract, Load, Transform) processes, API integrations (e.g., REST, GraphQL), and streaming data technologies. These pipelines will source data from diverse operational and monitoring tools (e.g., SolarWinds, cloud-native monitoring platforms, System Center) for ingestion into a centralized data platform (e.g., data warehouse, data lake, or data fabric).
- Perform comprehensive performance, functionality, and validation testing for current and emerging technologies. This includes documenting all methodologies, results, and findings for areas such as:
  - Cloud-Native Architectures (e.g., Containerization with Docker/Kubernetes)
  - Virtualization (Server and Desktop) and Virtual Desktop Infrastructure (VDI)
  - Desktop-as-a-Service (DaaS) solutions
  - Software-Defined Networking (SD-WAN) and Secure Access Service Edge (SASE)
  - Data storage optimization and protection technologies
- Conduct trade studies and analyses of alternatives (AoA) by comparing technology vendors and solutions to inform strategic decisions and define future technology roadmaps.
- Evaluate emerging technologies that can enhance organizational efficiency, such as Generative AI, hyper-automation, and Zero Trust frameworks, and provide well-researched recommendations for adoption.
- Develop solution-driven architectures that align with organizational goals and technical standards.
- Analyze and review proposed technical solutions (both internal and external) to ensure technical merit and compliance with organizational standards, as well as published industry and government best practices.
- Provide information regarding all systems operating within the HUD computing environment for inclusion in the One-HUD Systems Inventory.

#### **C.4.16 Operations and Maintenance (O&M)**

The Contractor shall operate, repair, and maintain systems, applications, and IT environments supporting various organizational requirements. Environments requiring O&M tasks may include pre-production, production, test, training, disaster recovery/fail over, or any other combination of IT accounts.

O&M services shall include, but are not limited to:

- Preventive and scheduled maintenance.
- Activities to retain or restore systems (e.g. testing, measurements, replacements, adjustments).
- Routine work required to maintain and/or enhance IT systems.
- Provide software, infrastructure, platform, telecommunications, and Storage-as-a-Service.
- Recommendations for best practice in requirements analysis, planning, design, deployment, and ongoing operations management.
- Technical support for software, infrastructure, platform, telecommunications, and storage services, which may be provided via subscription or other means.

The Contractor shall also maintain a current and up-to-date library of all operational documentation, including:

- Logs of operational events.
- Maintenance of operational monitoring and management tools.
- Operational scripts and procedures.

Specific O&M system tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.16.1 Systems/Network Administration**

The Contractor shall provide comprehensive support for IT hardware and operating systems, including but not limited to:

- Installation of software.
- Monitoring and adjusting system performance.
- Applying patches, security updates, and service packs.
- Repairing and upgrading IT hardware.
- Monitoring system resources (e.g. processor, memory and disk utilization) using automated monitoring tools.
- Analyze system logs and create system backups.
- Scheduling and tape allocation.
- Establishing and maintaining access authorizations.
- Performing installations, upgrades, or replacements as required.



Security Requirements: All default software passwords shall be changed prior to moving systems into a production environment.

#### **C.4.16.2 Application Support**

The Contractor shall provide code-level and operational support for applications, scripts, and middleware software, including but limited to:

- Code review, debugging, and patching.
- Error correction and defect repair.
- Performance monitoring.
- Training for required personnel.
- Configuring and installing upgrades/patches per maintenance agreements using change and release management processes

Specific application support tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.16.3 Hardware Support**

The Contractor shall install, configure, patch, repair, upgrade, replace or remove hardware systems, components and operating systems.

Security Requirement: All default hardware passwords shall be changed prior to moving systems into a production environment. Specific hardware support tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.16.4 Disaster Recovery (DR) and Continuity of Operations (COOP)**

The Contractor shall provide services related to all methodologies for disaster recovery and business continuity. Services shall cover scenarios ranging from partial loss of function or data to worst-case events involving complete IT enterprise failure due to natural disasters, man-made incidents, or IT failures. Services may be required from initial disaster declaration through full recovery of business processes. Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.16.5 Capacity and Availability Planning and Management**

The Contractor shall perform analysis and services to ensure that IT capacity meets current and future business requirements in a cost-effective manner. Activities include but not limited to:

- Monitoring historical utilization patterns.
- Forecasting based on emerging requirements.
- Ensuring IT service availability to support business operations.

#### **C.4.16.6 Service Desk/Help Desk/Call Center Support**

The Contractor shall provide full-service support for Service Desk, Help Desk, and Call Center operations as specified in individual TOs.

Responsibilities include but not limited to:

- Supporting Service Operation processes (e.g., incident management, event management, request fulfillment, access management, problem management, etc.).
- Operating within ITSM frameworks adopted by the organization.
- Meeting service-level requirements and metrics defined in each TOs.
- Providing onsite or remote system administration and operational support.
- Installing new software releases at supported locations/facilities/sites.
- Performing individual computer and peripheral maintenance and desk-side-services.

#### **C.4.16.7 Asset Management**

The Contractor shall provide asset management support, including inventory and utilization tracking for software and hardware assets as specified in individual TOs.

#### **C.4.16.8 License Maintenance**

The Contractor shall acquire, manage, and maintain licenses and/or commercial maintenance agreements for use on all proprietary and commercial software as appropriate and specified in individual TOs.

#### **C.4.16.9 Database and Data Warehouse Administration**

The Contractor shall provide comprehensive data management services, including:

- Logical and physical modeling and design/redesign.
- Installation, administration, tailoring, tuning, troubleshooting, integrating, patching, upgrading, and reporting.
- COOP, backup/recovery, archiving, encryption, and encryption key management.
- Development, maintenance, optimization, transition, and decommissioning of Extract Transform Load (ETL) capabilities and scripts.
- De-personalization of data and data protection procedure development.
- Data mining and Business Intelligence (BI) support, including product recommendation, selection, implementation, dashboard and report development, BI strategies, decision support, and data/report distribution.
- Enterprise data capture, curation, storage, search, processing, sharing, transfer, analysis, and virtualization.
- Near real-time analytics, including unstructured and structured, large and complex data at enterprise scale.
- Meet interoperability requirements at the Federal, state, and local levels.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.16.10 Data Center Administration**

The Contractor shall provide operations and maintenance for data centers, including:

- Preventive maintenance schedules.
- Emergency and corrective services.
- Coordination and tracking of service visits.
- Physical site inspections.
- Invoice and service report reviews.
- Resolution of service issues.

Physical operations and maintenance (O&M) including but not limited to:

- Above and below floor cleaning.
- Uninterruptible Power Supply (UPS) and battery maintenance.
- Power distribution inspections and corrections.
- Generator and automatic transfer switch maintenance.
- Fire suppression and detection equipment servicing.
- Heating, Ventilation, and Air Conditioning (HVAC) and environmental controls.

Additional responsibilities:

- Data center planning and design.
- Power and cooling analysis.
- Feasibility studies and risk assessments.
- Site selection and energy usage assessments.
- Virtualization strategies and optimization evaluations.
- Business continuity and disaster recovery planning.
- Relocation or consolidation planning.
- IT technology roadmap development.

The Contractor shall also provide services related to Software-Defined Data Center (SDDC), including topologies that extracts, pools and automates compute, network, and storage resources across multiple data centers. Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17 Cybersecurity**

The Contractor shall define and deliver strategic, operational and process aspects of cybersecurity solutions. The Contractor shall ensure the security of networks, data, information, and systems in accordance with organizational Standard Operating Procedures (SOPs), conditions, applicable laws, regulations, and policies. The Contractor shall comply with all

HUD information security requirements, including the protection of Personal Identification Information (PII) and Protected Health Information (PHI). Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17.1 Governance, Risk, Compliance, and Security Management**

The Contractor shall provide expert consultation and technical assistance for agency-level Authorization and Accreditation (A&A) processes, ensuring compliance with:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF).
- Applicable control families from NIST SP 800-53.

The Contractor shall:

- Support A&A activities throughout the system developmental lifecycle, including risk assessments, vulnerability scanning, and security control integration.
- Execute independent audits, continuous monitoring, and compliance reporting to confirm regulatory adherence and mitigate identified deficiencies.
- Assist with incident response planning, security training, and documentation oversight to strengthen cybersecurity resilience.
- Facilitate prompt remediation of audit findings and advance the agency's information security maturity.

#### **C.4.17.2 FISMA/NIST Cybersecurity Framework Analysis, Design, and Planning**

The contractor shall design, develop, deploy, and sustain capabilities that:

- Continuously ingest, analyze, and interpret current and future versions of FISMA and NIST CSF documentation.
- Identify changes, deltas, and emerging requirements across statutes, standards, and guidance.
- Generate updated plans, architectures, and design patterns translating evolving requirements into actionable technical and operational solutions.
- Maintain analytic logic, data sources, and mapping frameworks, threat conditions, and technology maturity.
- Provide forward-looking analysis to anticipate future control expectations and integration pathways for advanced cybersecurity capabilities.
- Enable proactive modernization and defensible compliance through structured, repeatable, and auditable outputs.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17.3 Information Assurance (IA)**

The Contractor shall identify, mitigate and resolve IA issues. Responsibilities shall include:

- Developing or contributing to guidelines, plans, policies, analyses requiring IA expertise
- Supporting assessments, monitoring, maintaining, reviewing and processing, A&A, accreditation/certification, Program Protection Plan (PPP) evaluation, and other cybersecurity related activities and mandates.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17.4 System Design/Sustainment**

The Contractor shall deliver comprehensive system design, development, deployment, and sustainment services for network security, including but not limited to:

- Architecture reviews, vulnerability assessments, and integration of NIST SP 800-53 controls.
- Regular updates, performance monitoring, and lifecycle management to maintain system integrity and compliance with federal standards.
- Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.17.4.1 Network and Transport Security**

The Contractor shall implement and sustain:

- Encryption protocols and secure data transmission mechanisms.
- Intrusion detection and prevention systems to safeguard information in transit across networks.
- Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.17.4.2 Endpoint and Platform Security**

The Contractor shall design, develop, deploy, and sustain endpoint and platform security for appliances, compute, storage, and related IT infrastructure. Activities include but not limited to:

- Hardening, patch management, and configuration baseline enforcement.
- Integration of NIST SP 800-53 control families (e.g. SI (System and Information Integrity), CM (Configuration Management), and SC (System and Communications Protection)).
- Implementation of access controls, encryption at rest, monitoring, and resilience features.

- Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17.4.3 Application & Software Supply Chain Security**

The Contractor shall provide expert system design, development, deployment, and sustainment services for application and software supply chain security. These services shall ensure compliance with:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF).
- NIST SP 800-171 Revision 3 requirements for protecting Controlled Unclassified Information (CUI).
- Relevant NIST SP 800-53 control families including:
  - SA (System and Services Acquisition).
  - SR (Supply Chain Risk Management).

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.17.4.3.1 Application & Software Supply Chain Security Core Requirements**

The Contractor shall:

- Incorporate secure software development practices throughout the application lifecycle.
- Perform third-party risk assessments and maintain provenance tracking for all components.
- Generate, review, and maintain Software Bills of Materials (SBOMs) in machine-readable formats.
- Conduct vendor due diligence and continuous monitoring of third-party software and services in accordance with NIST SP 800-161 guidelines.
- Integrate SBOM data to identify vulnerabilities and address weaknesses in the supply chain.
- Implement and sustain:
  - Automated vulnerability scanning.
  - Dependency management.
  - SBOM-informed incident response processes for ongoing threat visibility and rapid remediation.

These measures shall enable proactive mitigation of supply chain risks, compliance with FISMA and NIS SP 800-171, and advancement toward higher cybersecurity maturity.

#### **C.4.17.4.4 Sustainment Activities**

The Contractor shall:

- Maintain application security designs, encompassing secure coding standards, performance of static and dynamic analysis, and enforcement of configuration hardening.
- Support software supply chain security through lifecycle management, including:
  - Regular SBOM updates.
  - Verification of third-party patches and components.
  - Evaluation of emerging risks per NIST SP 800-161 and SR family requirements.
- Integrate secure-by-design principles, encryption of sensitive data, access controls, and integrity checks for third party components.
- Conduct periodic:
  - Risk assessments.
  - SBOM audits.
  - Third-party supply chain reviews to validate sustained effectiveness.
- Ensure these services:
  - Proactive risk mitigation.
  - Regulatory compliance with NIST SP 800-171.
  - Resilient operation of critical applications and software ecosystems.

#### **C.4.17.4.5 Data Security and Information Protection (DSaIP)**

The Contractor shall provide expert design, development, deployment, and sustainment services for data security and information protection, ensuring comprehensive safeguards for sensitive data – in including Controlled Unclassified Information (CUI) – in alignment with:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF).
- NIST SP 800-171 Revision 3 requirements.
- Relevant NIST SP 800-53 control families, including:
  - SC (System and Communications Protection).
  - MP (Media Protection).
  - SI (System and Information Integrity).

##### **C.4.17.4.5.1 DSaIP Core Requirements**

The Contractor shall:

- Implement and maintain encryption mechanisms for data at rest and in transit.
- Enforce access controls and secure storage practices to safeguard confidentiality, integrity, and availability throughout the data lifecycle.
- Apply secure data handling measures, including:
  - Data labeling.
  - Dissemination controls.

- Integrity verification during creation, processing, and output activities.

#### **C.4.17.4.5.2 Monitoring and Risk Management**

The Contractor shall:

- Conduct regular data classification reviews and vulnerability assessments of data repositories.
- Implement continuous monitoring capabilities to detect and respond to anomalies or breaches affecting data integrity.
- Integrate and operate automated monitoring and risk-management tools for:
  - Audit logging.
  - Backup management.
  - Compliance reporting to support FISMA oversight and NIST SP 800-171 self-assessments or third-party evaluations.

#### **C.4.17.4.5.3 DSaIP Outcome**

Through these sustained efforts, the Contractors shall enable:

- Proactive risk mitigation.
- Regulatory compliance.
- Resilient protection of organizational data assets and information workflows.

#### **C.4.17.4.6 Cloud and Virtualized Environment Security (CaVES)**

The Contractor shall provide expert design, development, deployment, and sustainment services for cloud and virtualized environment security. These services shall ensure compliance with:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF).
- NIST SP 800-171 Revision 3 requirements for protecting Controlled Unclassified Information (CUI).
- Applicable NIST SP 800-53 control families, including, but not limited to:
  - SC (System and Communications Protection).
  - CM (Configuration Management).
  - SA (System and Services Acquisition).

##### **C.4.17.4.6.1 CaVES Core Requirements**

The contractor shall:

- Implement, operate, and maintain secure cloud architectures, including:
  - Hypervisor hardening.
  - Virtual machine isolation.



- Container orchestration security.
- Mitigate risks in multi-tenant and virtualized infrastructures through robust configuration and access controls.

#### **C.4.17.4.6.2 CaVES Sustainment Activities**

The Contractor shall:

- Conduct continuous security monitoring, automated configuration management, patch management, and vulnerability scanning tailored to:
  - Cloud service models (IaaS, PaaS, SaaS).
  - Virtualization technologies.
- Align practices with NIST SP 800-144 guidelines and FedRAMP-approved controls.
- Enforce:
  - Encryption for data at rest and in transit.
  - Identity and Access Management (IAM) with least privilege principles.
  - Network segmentation to prevent lateral movement and unauthorized access.

#### **C.4.17.4.6.3 CaVES Risk and Compliance Management**

The Contractor shall:

- Perform regular risk assessments and compliance audits.
- Integrate incident response capabilities to ensure resilience and meet FISMA reporting obligations.

#### **C.4.17.4.6.4 CaVES Outcome**

The Contractor shall enable:

- Proactive threat mitigation.
- Regulatory compliance.
- Sustained operational security for cloud and virtualized environments.

#### **C.4.17.4.7 Security Lifecycle**

The Contractor shall provide expert design, development, deployment, and sustainment services for secure system and software lifecycle management within an enterprise DevSecOps platform. These services shall ensure compliance:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF).
- NIST SP 800-171 Revision 3 requirements for protecting Controlled Unclassified Information (CUI).
- NIST SP 800-53 relevant control families, including:

- o SA (System and Services Acquisition).
- o SR (Supply Chain Risk Management).
- o SI (System and Information Integrity).

#### **C.4.17.4.7.1 Security Lifecycle Core Requirements**

The Contractor shall:

- Design, implement and operate a full enterprise DevSecOps platform with:
  - o Centralized, multi-tenant architectures that enable multiple simultaneous projects and development streams.
  - o Shared security tooling, governance, and automated CI/CD pipelines for consistent, scalable delivery.
- Implement enterprise-level orchestration, shared repositories, cross-project dependency management, and reference interconnects to:
  - o Ensure interoperability.
  - o Prevent silos.
  - o Facilitate seamless merging of artifacts from parallel efforts.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17.4.7.2 Security Lifecycle Integration**

The Contractor shall:

- Enforce automated security testing with pipelines, including:
  - o Static Application Security Testing (SAST).
  - o Dynamic Application Security Testing (DAST).
  - o Software Composition Analysis.
  - o Interactive testing for early vulnerability detection across all concurrent project.

Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.17.4.7.3 Security Lifecycle Distribution**

The Contractor shall:

- Sustain secure distribution processes through:
  - o Signed artifacts.
  - o Encrypted channels.
  - o Provenance verification using Software Bills of Materials (SBOMs).
  - o Controlled enterprise release workflows to mitigate deployment and update risks in multi-project environments.

#### **C.4.17.4.7.4 Security Lifecycle Outcome**

Through these efforts, the Contractor shall enable:

- Proactive threat mitigation.
- Regulatory compliance with NIST SP 800-171.
- Sustained operational security across the full enterprise DevSecOps lifecycle supporting multiple simultaneous projects.

#### **C.4.17.4.8 Policy-as-Code**

The Contractor shall design, develop, deploy, and sustain policy-as-code solutions that enable automated enforcements of cybersecurity and privacy requirements. These policies shall be machine-readable and aligned with:

- Applicable NIST SP 800-53 control families.
- Agency governance requirements.

##### **C.4.17.4.8.1 Policy-as-Code Core Requirements**

The Contractor shall:

- Implement policy-as-code frameworks that support:
  - Automated enforcement.
  - Validation.
  - Continuous compliance across cloud, on-premises, and hybrid environments.
  - Ensure integration with existing legacy systems while providing extensibility for advanced services, including but not limited to:
    - Automated authorization.
    - Continuous monitoring capabilities.

##### **C.4.17.4.8.2 Policy-as-Code Management**

The Contractor shall:

- Maintain policy logic that is:
  - Version-controlled.
  - Auditable.
  - Testable within modern DevSecOps pipelines to support secure change management.
- Ensure policies remain current with:
  - Evolving control requirements.
  - Threat conditions.
  - Technology platforms.

#### **C.4.17.4.8.3 Policy-as-Code Outcome**

The Contractor shall enable:

- Automated compliance verification.
- Scalable governance across diverse environments.
- Reduced risk through consistent, enforceable security and privacy policies.

#### **C.4.17.4.9 Resilience, Continuity, and Recovery (RCR)**

The Contractor shall provide expert design, development, deployment, and sustainment services for resilience, continuity, and recovery. These services shall ensure compliance with:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF) - Recover function.
- NIST SP 800-171 Revision 3 requirements.
- NIST SP 800-53 relevant control families, including:
  - CP (Contingency Planning).
  - IR (Incident Response).
  - SI (System and Information Integrity).

##### **C.4.17.4.9.1 RCR Core Requirements**

The Contractor shall:

- Define and implement:
  - Recovery Time Objective (RTO): Maximum acceptable downtime for system restoration.
  - Recovery Point Objective (RPO): Maximum tolerable data loss.
- Tailor RTO and RPO metrics to mission/business impact analyses and system criticality.

##### **C.4.17.4.9.2 RCR Recovery Strategies**

The Contractor shall support a range of recovery approaches, including but not limited to:

- Cold sites: Basic infrastructure requiring significant setup time.
- Warm sites: Partially equipped with moderate activation speed.
- Hot sites: Fully mirrored environments enabling near-instant failover.

To meet specified RTO and RPO targets, the Contractor shall integrate:

- Full and incremental backups.
- Real-time replication.
- Electronic vaulting.
- Offsite storage and automated failover mechanisms.

#### **C.4.17.4.9.3 RCR Coordination and Testing**

The Contractor shall:

- Establish and sustain coordination between continuity/recovery processes and incident response activities.
- Develop joint planning artifacts, shared playbooks, and escalation procedures.
- Conduct integrated testing to ensure seamless transition from incident containment to full system recovery.

#### **C.4.17.4.9.4 RCR Outcome**

Through these efforts, the Contractor shall enable:

- Proactive risk mitigation.
- Regulatory compliance.
- Effective incident-to-recovery coordination.
- Resilient operational continuity for critical systems and data.

#### **C.4.17.4.10 Emerging and Advanced Capability Cybersecurity Solutions (EaACCS)**

The Contractor shall provide expert design, development, deployment, and sustainment services for advanced cybersecurity solutions tailored to protect the financial integrity and operational components of HUD programs. These services shall ensure compliance with:

- Federal Information Security Modernization Act (FISMA).
- NIST Cybersecurity Framework (CSF).
- NIST SP 800-171 Revision 3 requirements for Controlled Unclassified Information (CUI).
- NIST SP 800-53 relevant control families, including:
  - o SI (System and Information Integrity).
  - o SC (System and Communications Protection).
  - o IR (Incident Response).

##### **C.4.17.4.10.1 EaACCS Core Capabilities**

The Contractor shall:

- Implement and maintain Security Orchestration, Automation, and Response (SOAR) platforms to automate threat detection, incident triage, and response workflows for HUD's housing finance systems, grant management, and data repositories.
- Integrate post-quantum cryptography and quantum-resistant protections using NIST-standardized algorithms (e.g. FIPS 203, 204, and 205) to safeguard encryption keys, digital signatures, and sensitive data against future quantum threats.

- Conduct cryptographic inventories and migration planning to future-proof financial transactions and housing program data.
- Deploy AI-driven security capabilities for:
  - Anomaly detection.
  - Predictive threat intelligence.
  - Automated risk scoring.
- Implement AI-resistant measures, including adversarial training, model hardening, and secure AI governance to counter AI-assisted attacks or model poisoning.
- Incorporate advanced technologies such as:
  - Zero-trust architectures.
  - Behavioral analytics.
  - Continuous monitoring for resilience across HUD's distributed IT environment.

#### **C.4.17.4.10.2 EaACCS Sustainment and Testing**

The Contractor shall:

- Provide ongoing sustainment through:
  - Regular updates.
  - Performance tuning.
  - Integration testing to adapt to evolving threats.
- Conduct periodic:
  - Vulnerability assessments.
  - Penetration testing.
  - Red-team exercises focused on quantum-vulnerable and AI-exploitable vectors .
- Validate and document the effectiveness of:
  - SOAR orchestration.
  - Post-quantum encryption deployments.
  - AI security controls.

#### **C.4.17.4.10.3 EaACCS Financial Integrity and Compliance**

The Contractor shall:

- Prioritize safeguards for critical HUD data flows, including but not limited to:
  - Mortgage insurance processing.
  - Rental assistance.
  - Grant disbursements.
- Support compliance reporting and audit readiness under FISMA and related federal mandates.

#### **C.4.17.4.10.4 EaACCS Collaboration and Knowledge Transfer**

The Contractor shall:

- Collaborate with HUD stakeholders to develop:
  - Customized threat-hunting strategies.
  - Incident response playbooks.
  - Recovery coordination plans.
- Provide training and knowledge transfer on advanced technologies to ensure long-term operational self-sufficiency.

#### **C.4.17.4.10.5 EaACCS Outcome**

Through these efforts, the Contractor shall enable:

- Elevated cybersecurity maturity.
- Robust protection against sophisticated adversaries.
- Sustained trust in HUD's delivery of affordable housing solutions nationwide.

#### **C.4.17.5 Assessment and Authorization (A&A)**

The Contractor shall obtain commercial and/or agency specific certifications and authorizations for new or modified systems, applications, designs, equipment, or installations in accordance with applicable agency standards and individual TOs requirements.

##### **C.4.17.5.1 A&A Core Activities**

The Contractor shall:

- Perform security authorization and assessment activities, including certifications and comprehensive assessments of management, operational, and technical security controls within information systems.
- Determine and document the extent to which the security controls are:
  - Implemented correctly.
  - Operating as intended.
  - Producing the desired level of protection.

These activities shall ensure compliance with applicable federal and organizational security requirements and support authorization decisions for system operation.

#### **C.4.17.6 Cybersecurity Operational Support (CyOS)**

The Contractor shall provide comprehensive operational support for cybersecurity services at both local and enterprise levels. These services shall ensure continuous protection, monitoring, and compliance across the agency's IT environment.

#### **C.4.17.6.1 CyOS Core Capabilities**

The Contractor shall deliver, but is not limited to:

- 24/7 Cyber Watch Operations.
- Intrusion Detection and Prevention System (IDS/IPS).
- Network discovery and security device monitoring.
- Penetration testing and compliance scanning.
- Vulnerability Scanning Service (VSS) and vulnerability management assistance.
- Patch management and telemetry/log ingestion.
- Endpoint protection and cybersecurity incident response.
- Digital forensics and enterprise network defense.
- Cyber threat and business intelligence.
- Insider threat analysis and Identity and Credential Access Management (ICAM).
- Development and analysis of operational and cybersecurity metrics.
- Information security continuous monitoring.
- Cybersecurity posture verification and validation.
- Reporting to external agencies and cybersecurity exercise coordination.

#### **C.4.17.6.2 CyOS Operational Support Activities**

The Contractor shall:

- Provide cybersecurity analytics support, incident identification, handling, and remediation.
- Conduct digital forensics, threat hunting, and intelligence gathering/sharing.
- Perform correlation and analysis of threat data and research emerging threats.
- Assess cybersecurity tools for effectiveness and ensure compliance with federal mandate and directives.

#### **C.4.17.6.3 CyOS Enterprise-Level Responsibilities**

The Contractor shall:

- Participate in security functions to ensure system integrity and availability, including:
  - Security safeguard reviews.
  - Audits.
  - Reporting suspected violations.
  - Securing system environments.
  - Monitoring and responding to alerts.
- Perform enterprise-wide analysis of centralized log management and data correlation.



#### **C.4.17.6.4 CyOS Specialized Services – Cyber Orders**

The Contractor shall design and execute operational frameworks that translate executive direction into coordinated, repeatable defensive campaigns. Responsibilities include, but not limited to:

- Developing and deploying tooling, workflows, and metrics for campaign planning, execution, and assessment.
- Ensuring adaptability to evolving threats, technologies, and mission requirements.

#### **C.4.17.6.5 CyOS Specialized Services - Cyber Intelligence Fusion**

The Contractor shall design, integrate, and operate systems that aggregate and correlate internal telemetry, external intelligence, and partner information. Activities include, but not limited to:

- Developing analytic processes and platforms for timely, risk-informed decisions.
- Collecting and operationalizing indicators or compromise.
- Creating threat actor profiles and trend analyses.
- Monitoring sector-specific threats, brand reputation, and VIP impersonation risks.
- Maintaining data integrity, analytic effectiveness, and trusted information-sharing capabilities.

#### **C.4.17.6.6 CyOS Digital Forensics Incident Response, Threat Hunting, and Active Defense**

The Contractor shall:

- Design, deploy, and operate persistent capabilities supporting investigation, detection, and adversary emulation.
- Develop forensic readiness and proactive hunting methodologies.
- Conduct blue, red, and purple team exercises to validate defenses.
- Conduct forensic investigations, preserve and handle evidence, prepare after-action reports, and integrate lessons learned into detection content.
- Ensure continuous skill development, tool effectiveness, and integration of lessons learned.
- Leverage findings to improve the agency's defensive capabilities and build detection content to simplify future detections of similar threats.

#### **C.4.17.6.7 CyOS Cyber Watch and Intrusion Detection**

The Contractor shall:

- Provide (24/7x365) Cyber Watch operations, security event monitoring, and intrusion detection.
- Perform initial triage and escalation of security events.
- Provide 24/7 monitoring of security dashboards, tool health, mailboxes, and calls.

- Serve as the initial intake and triage of security events that may escalate further.
- Collect, track, automate and report FISMA IR metrics reporting.
- Maintain playbooks, SOPs, and training materials.

#### **C.4.17.6.8 CyOS Secure Cloud/Commercial Solutions**

The Contractor shall:

- Design, deploy, operate, and sustain capabilities for secure cloud or commercial services operations.
- Integrate solutions using standardized architectures, identity-centric controls, and continuous monitoring.
- Ensure security capabilities evolves alongside commercial services without increasing operational risk.

#### **C.4.17.6.9 CyOS Enterprise Operations**

The Contractor shall:

- Develop, implement and maintain a unified operating model integrating governance, workforce, processes, and technology.
- Deploy and supporting systems for consistent execution across all environments.
- Preserve operational coherence, scalability, and executive visibility.
- Support enterprise-wide cybersecurity operations (e.g. data center operations, customer experience, internet of things, mobile device management, applications management, release management).

#### **C.4.17.6.10 CyOS Vulnerability and Exposure Management**

The Contractor shall:

- Design, deploy, operate, and sustain continuous discovery, assessment, and remediation capabilities.
- Ensure compliance with Executive Orders, Binding Operational Directives, Emergency Directives, and the CISA Known Exploited Vulnerabilities (KEV) catalog.
- Implement threat-informed prioritization and automated remediation workflows.

#### **C.4.17.6.11 CyOS Insider and Behavior Risk Management**

The Contractor shall:

- Design, deploy, and operate capabilities to identify and mitigate risks from misuse, error, or compromised trust.
- Integrate technical monitoring, behavioral analysis, and governance controls in compliance with applicable laws and policies.

#### **C.4.17.6.12 CyOS Red Team Operations**

The Contractor shall:

- Conduct penetration testing, adversary simulation, Zero-Trust validation, and purple team exercises.
- Develop Rules of Engagement (ROE) and provide detailed findings, attack paths, and prioritized remediation plans.
- Test defenses against advanced threat actors using modern attack techniques.
- Test the agency's entire security posture against highly sophisticated threat actors to evaluate the effectiveness of the organization's detection, response, and resilience capabilities (e.g. evading detection, lateral movement, privilege escalation, and social engineering using modern attack techniques).
- Establish framework for target selection based on risk factors and align testing efforts to the agency's needs.

#### **C.4.17.6.13 CyOS Security Engineering**

The Contractor shall:

- Design, implement, and maintain security capabilities for hybrid infrastructure (on-premises, cloud, SaaS).
- Maintain a robust suite of security tools including:
  - Security Information and Event Management (SIEM).
  - Security Orchestration, Automation, and Response (SOAR).
  - Endpoint Detect and Response (EDR).
  - User and Entity Behavioral Analytics (UEBA).
  - Data Loss Protection (DLP).
  - Web Application Firewall (WAF).
  - Full Packet Capture (PCAP).
  - Network Detect and Response (NDR).
  - Cloud Security.
  - E-mail security.
  - Phishing testing.
- Develop and maintain SOPs, guides, and other training materials.
- Perform security capability gap analysis and recommend innovative solutions.

#### **C.4.17.7 Identity, Credential, and Access Management (ICAM)**

The contractor shall provide design, development and deployment, and sustainment services for advanced identity, credential, and access management (ICAM) capabilities. These services shall ensure that only properly authenticated and authorized users gain access to agency systems and data.

#### **C.4.17.7.1 ICAM Core Requirements**

The Contractor shall:

- Design, implement and deploy identity architectures that enforce:
  - Strong identity proofing.
  - Credential lifecycle management.
  - Continuous validation of identity integrity.
- Implement, operate and maintain Role-Based Access Control (RBAC) models, including emergent control models to:
  - Align system access with job functions.
  - Enforce least-privilege principles.
  - Support documented authorities.

#### **C.4.17.7.2 ICAM Access Governance**

The Contractor shall:

- Develop, implement and sustain:
  - Authoritative role definitions.
  - Access mappings.
  - Approval workflows to prevent privilege accumulation and unauthorized access.
- Ensure identity integrity services maintain:
  - Accuracy.
  - Tamper-resistant.
  - Traceable identities, credentials, and attributes across all environments ( cloud, on-premises, and contractor-operated systems).

#### **C.4.17.7.3 ICAM Sustainment Activities**

The Contractor shall:

- Conduct continuous monitoring, periodic access reviews.
- Identify, remediate document orphaned, excessive, dormant or anomalous privileges.
- Maintain compliance with applicable federal stands and agency policies.

#### **C.4.17.7.4 ICAM Outcome**

The Contractor shall

- Defensible and resilient identity control foundation.
- Mission continuity and auditability.
- Executive confidence in secure access management.

#### **C.4.18 Privacy**

##### **C.4.18.1 Privacy Policies, Plans, and Governance**

The Contractor shall design, implement, and sustain privacy governance artifacts that integrate with enterprise cybersecurity and risk management frameworks. Specific privacy related tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

Responsibilities include but not limited to:

- Developing, updating and maintaining privacy policies, plans, and procedures that govern data handling, system operation, and lifecycle management.
- Ensuring privacy requirements are consistently applied across systems, contractors, and operational environments.
- Establishing governance mechanisms shall support accountability, auditability, and executive oversight.

The Contractor shall provide NIST privacy control mapping and alignment services including but not limited to:

- Identifying, documenting, and maintaining traceability between privacy requirements and applicable NIST Privacy Framework functions and NIST SP 800-53 privacy controls.
- Developing and sustaining control mappings that demonstrate how privacy policies, processes, and technical mechanisms are implemented across systems and services.
- Ensuring privacy mappings remain current as systems, controls, and regulatory requirements evolve.

##### **C.4.18.2 Privacy Monitoring and Assessment**

The Contractor shall:

- Continuously evaluate compliance with privacy requirements and identify emerging risks.
- Conduct Privacy Impact Assessments (PIAs), threshold analyses, and ongoing monitoring activities aligned with system changes and operational use.
- Document, track, and report assessment results to support risk-based decision-making.
- Sustain privacy posture as systems, data, and threats evolve.

### **C.4.18.3 Privacy-as-Code**

The Contractor shall:

- Integrate privacy requirements directly into system design, development, and deployment processes.
- Implement machine-readable privacy controls, APIs, and automated testing to enforce:
  - Data minimization.
  - Purpose limitation.
  - Access restrictions.
- Ensure privacy controls remain effective and verifiable throughout the system lifecycle.

### **C.4.18.4 Privacy Training and Awareness**

The Contractor shall:

- Design, deliver, and maintain role-based training plans and curricula tailored to workforce, contractor, and leadership responsibilities.
- Measure training effectiveness and update content to address emerging risks and regulatory changes.
- Sustain consistent awareness and accountability across the enterprise.

### **C.4.18.5 Privacy Incident Response**

The Contractor shall:

- Design, implement, and sustain capabilities to identify, assess, contain, and remediate privacy incidents and breaches.
- Develop incident response plans, workflows, and coordination mechanisms aligned with cybersecurity incident response processes.
- Support timely notification, investigation, and corrective action in accordance with applicable laws and policies.
- Ensure readiness through testing, exercises, and continuous improvement.

## **C.4.19 Records Management**

### **C.4.19.1 Records Governance and Accountability**

The Contractor shall design, implementation, and sustain enterprise records governance framework that establishes clear roles, responsibilities, and authorities for records management across all organizational components. Governance mechanisms shall ensure accountability, oversight, and alignment with statutory and regulatory requirements, including but not limited to:

- 36 CFR Chapter XII.
- OMB Circular A-130 for federal information governance and accountability.

Specific record management related tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

#### **C.4.19.2 Records Management Lifecycle Management**

The Contractor shall provide end-to-end records lifecycle management services, encompassing creation or receipt through final disposition. Responsibilities include but not limited to:

- Designing and operating processes to capture, maintain, and dispose of records in accordance with approved schedules.
- Applying lifecycle management consistently across all environments and record types.

This requirement aligns with 36 CFR §§1220–1228 and OMB Circular A-130 lifecycle management principles.

#### **C.4.19.3 Records Classification**

The Contractor shall implement electronic records capture and classification services including but not limited to:

- Automated identification, capture, and classification of electronic records at the point of creation or receipt.
- Accurate application of metadata, record categories, and security markings.
- Support both structured and unstructured content.

Aligned with 36 CFR §1222 and OMB Circular A-130 requirements for electronic records management requirements.

#### **C.4.19.4 Retention, Scheduling and Disposition**

The Contractor shall:

- Implement and operate approved records retention schedules.
- Ensure timely, auditable, and legally defensible disposition of records.
- Track, document, and retain evidence of disposition actions for oversight and risk management.

Aligned with 36 CFR §§1225–1226 and OMB Circular A-130 disposition controls.

#### **C.4.19.5 Permanent Records Management**

The Contractor shall:

- Identify, preserve, and prepare permanent records for transfer to the National Archives (NARA).

- Ensure compliance with format, metadata, and quality requirements.
- Maintain transfer readiness continuously, not event-driven.

Aligned with 36 CFR § 235 and OMB Circular A-130 archival transfer requirements.

#### **C.4.19.6 Electronic Records Management**

The Contractor shall:

- Capture and manage records generated through email, chat, collaboration platforms.
- Implement automated capture mechanisms to minimize user burden while ensuring completeness.
- Manage electronic records according to approved schedules and policies.

Align with 36 CFR §1236 and OMB Circular A-130 electronic messaging requirements.

#### **C.4.19.7 Line of Business Services**

The Contractor shall:

- Ensure records embedded within operational systems are properly identified and managed.
- Integrate records management controls directly into mission and financial systems.
- Preserve security and integrity throughout system operation.

Aligned with 36 CFR §1222.34 and OMB Circular A-130 system design requirements.

#### **C.4.19.8 Media Management**

The Contractor shall:

- Design, implement and operate secure storage solutions that protect records from loss, corruption, or unauthorized alteration.
- Manage storage media to ensure durability, accessibility, and integrity.
- Implement preservation strategies for format obsolescence and migration needs.

Aligned with 36 CFR §1236.28 and OMB Circular A-130 information protection requirements.

#### **C.4.19.9 Search and Retrieval Services**

The Contractor shall:

- Provide timely and accurate discovery of records to authorized users.
- Implement indexing, search, and retrieval capabilities supporting operational, legal, and oversight needs.



- Enforce access controls to ensure authorized retrieval only.

Aligned with 36 CFR §1222.18 and OMB Circular A-130 access and usability requirements.

#### **C.4.19.10 Legal support**

The Contractor shall:

- Provide legal-support services including legal hold, eDiscovery, and litigation support services.
- Place, manage, monitor and release legal holds across all relevant systems.
- Support discovery, preservation, and production of records for litigation and investigations.
- Suspend disposition during legal holds while maintaining chain of custody.

Aligned with 36 CFR §1226.18, 36 CFR §1222.32, NARA Subchapter B and OMB Circular A-130 legal and risk management requirements.

##### **C.4.19.10.1 Records Privacy**

The Contractor shall:

- Apply RBAC and privacy safeguards for sensitive or regulated records.
- Continuously monitor and mitigate privacy risks.

Aligned with 36 CFR Sections 1256.56, and OMB Circular A-130 privacy and access control requirements.

##### **C.4.19.10.2 Commercial and Cloud Services**

The Contractor shall:

- Ensure records managed by cloud, SaaS, and managed service platforms remain under enterprise control.
- Design, implement and enforce requirements for capture, retention, access, and disposition in cloud in cloud environments.
- Perform and document customer responsibilities within shared responsibility models.

Aligned with 36 CFR §1236.14 and OMB Circular A-130 cloud records management guidance.

##### **C.4.19.10.3 Disposition Assurance**

The Contractor shall:

- Verify disposition actions are executed correctly and defensibly.

- Implement controls and reporting that demonstrate compliance and reduce legal and operational risk.
- Identify, document and remediate exceptions.

Aligned with 36 CFR §1226.24 and OMB Circular A-130 risk management provisions.

#### **C.4.19.10.4 Workforce Enablement**

The Contractor shall:

- Design training, guidance, and tools to integrate records management into daily work practices.
- Develop and implement adoption strategies that minimize burden while reinforcing accountability.

Aligned with 36 CFR §1220.34 and OMB Circular A-130 workforce responsibility requirements.

#### **C.4.19.10.5 Records Modernization**

The Contractor shall:

- Integrate automation, analytics, and emerging technologies to improve efficiency and compliance.
- Implement advanced capabilities such as automated classification, workflow orchestration, and risk-based controls.
- Ensure modernization integrates with legacy systems while enabling future capabilities.

Aligned with 36 CFR §1236 and OMB Circular A-130 modernization and continuous improvement guidance.

#### **C.4.19.10.6 Records and NIST Cyber Security Framework Alignment**

The Contractor shall:

- Align records management services with NIST Special Publication 800-53 control families, including but not limited to:
  - o PL (Planning).
  - o PM (Program Management).
  - o RA (Risk Assessment).
  - o IA (Identification and Authentication).
  - o AC (Access Control).
  - o MP (Media Protection).
  - o CP (Contingency Planning).
  - o IR (Incident Response).
  - o AR (Authority and Responsibility).
  - o SR (Supply Chain Risk Management).

- Ensure records-management governance, lifecycle management, privacy protection, access controls, auditability, and disposition activities support confidentiality, integrity, availability, and accountability of federal records across legacy, cloud, and managed service environments.
- Maintain alignment as NIST SP 800-53 updates and as systems, threats, and operational requirements evolve.

### C.4.20 Customer Experience

#### C.4.20.1 Experience Design

The Contractor shall provide comprehensive support for the planning, management, and execution of customer experience (CX), digital experience (DX), user experience (UX), and employee experience (EX) initiatives across all phases of the design and implementation lifecycle. These initiatives shall include but not limited to:

- Reducing costs and improve service delivery efficiency.
- Enhancing customer satisfaction and usability.
- Ensuring compliance with the Government Service Delivery Improvement Act (GSDIA) and the 21st Century Integrated Digital Experience Act (21st Century IDEA).
- Aligning HUD services with customer needs and expectations.

The Contractor shall apply industry-standard experience design principles and best practices; assess customer and user feedback across multiple interaction channels (e.g., surveys, direct interactions, digital analytics, usability testing); and synthesize findings into actionable insights that inform design.

Deliverables for individual TOs may include but not limited to:

- **User Personas** – Evidence-based representations of key user types.
- **User Archetypes** – Generalized user profiles to inform design decisions.
- **User Flows and Journey Maps** – Visual representations of user tasks and interactions.
- **Site Maps and Content Structure Inventories** – Structural diagrams and inventories for HUD digital platforms.
- **Comparative Analyses** – Evaluations of alternative approaches, solutions, or design options.
- **Prototypes** – Low- to high- fidelity prototypes refined through testing and validation.
- **UX Expert and Heuristic Reviews** – Identification of usability and accessibility issues.
- **User Requirements Documentation** – Functional requirements based on user needs.
- **User Insights and Testing Reports** – Findings from user research and testing activities.
- **Research and User Testing Artifacts** – Documentation of methodologies, results, and lessons learned.

#### C.4.20.2 Digital Experience Transformation

The Contractor shall support modernization of HUD's websites and online systems to deliver simple, intuitive, and efficient digital experience for public and partner customers. Activities include but not limited to:

- Designing and implementing self-service capabilities (e.g., AI chatbots, dynamic FAQs, customer portals) to reduce contact center volume and costs.
- Consolidating and modernizing customer-facing web applications for a seamless digital experience across HUD programs and services.

Services for individual TOs may include but not limited to:

- **DX Analysis and Planning** - Assess digital experience ecosystem, identify inefficiencies, and recommend consolidation and modernization strategies.
- **Website Design** - Apply user-centered methods (usability testing, task testing, card sorting, and tree testing).
- **Website Development** - Provide technical expertise for Drupal content management system (CMS) and related technologies (PHP, HTML, CSS, JavaScript, and database technologies).
- **Web Application Design** - Develop UX and information architecture aligned U.S. Web Design System (USWDS), HUD branding, and federal mandates.
- **Web Application Development** - Support development, integration, and maintenance of HUD web applications.
- **Content and Data Migration** - Plan and execute content and data migration activities, including source assessment, sensitivity analysis, mapping, preparation, and migration support for new or modernized websites and web applications.
- **DX Governance** - Develop governance frameworks for sustainable website and web application management, content quality, and publishing efficiency.
- **Legislative Compliance** - Ensure digital services meet accessibility, security, user-centered, mobile-first, discoverable, branding requirements, data-driven, and optimized for search.

#### C.4.20.3 Contact Center / Assisted Support Consolidation and Modernization

The Contractor shall consolidate and modernize HUD's customer service operations by:

- Streamlining contact centers and assisted support channels.
- Optimizing inquiry management to reduce Tier II/III escalations.
- Developing and maintaining CRM Customer Engagement Center (CEC) and knowledge management systems.
- Integrating CCaaS and telephony systems with AI capabilities.
- Supporting onboarding and change management for program offices.

Deliverables for individual TOs may include but not limited to:

- **Current State Assessment** – Identify friction points, inefficiencies, and best practices; define success metrics and critical features; and engage HUD SMEs, agents, and supervisors.
- **Experience and Business Process Design** – Develop personas, use cases, workflows, IVR/agent/AI chatbot scripts, escalation paths, training materials, and standardized processes.
- **Requirements Gathering and Analysis** – Define requirements for CRM CEC, knowledge management, and CCaaS/telephony platforms; and provide recommendations.
- **Change Management** – Provide risk assessment, communications, user testing, training, piloting, and transition support.
- **Implementation and Management** – Support deployment, operations, and ongoing management of consolidated contact center platforms and systems.

#### C.4.20.4 Voice of the Customer / Customer Experience Analytics

The Contractor shall implement a Voice of the Customer (VoC) and Customer Experience (CX) measurement and analytics framework to inform service design, delivery, and continuous improvement of HUD's service channels.

Deliverables for individual TOs may include but not limited to:

- **VoC Data Collection and Analysis** – Collect, monitor, and analyze feedback across multiple channels using HUD's VoC platform.
- **VoC Capabilities Enablement** – Integrate of speech-to-text, text analytics, digital surveys, and analytics platforms (e.g., Google Analytics and Medallia); dashboard design and reporting enhancements.
- **CX Analytics** – Use qualitative and quantitative data to support CX/DX/UX modernization and compliance with federal mandates.
- **CX Insights and Reporting** – Synthesize insights with actionable recommendations to improve service delivery.
- **Legislative Compliance** – Support for required CX reporting, including OMB A-11 Section 280, GSDIA, and GPRA requirements.

#### C.4.20.5 Customer Engagement

The Contractor shall design, implement, and sustain proactive, inclusive, and coordinated engagement strategies aligned with HUD's CX strategy, VoC insights, and applicable legislative requirements. Deliverables for individual TOs may include but are not limited to:

- **Customer Engagement Strategy** – Define goals, audiences, channels, messaging principles, and success metrics.
- **Stakeholder Mapping and Engagement Planning** – Identify audience and develop tailored engagement plans with accessibility and feedback mechanisms.

- **Content and Communication Enablement** – Deliver plain-language, accessible, and consistent customer-facing content across all channels.
- **Proactive and Digital Engagement** – Implement lifecycle-based messaging, onboarding journeys, notifications, and self-service prompts.
- **Engagement Measurement and Optimization** – Use metrics, analytics, and continuous improvement using VoC and CX data.
- **Governance and Coordination** – Establish engagement governance and cross-office coordination to ensure a unified and sustainable customer experience.

#### C.4.21 Training

The Contractor shall identify training requirements, obtain or develop training programs, and conduct training for technologies, systems, applications and products at any stage of the lifecycle. This includes, but is not limited to:

- IT workforce development and competency-based training.
- Newly developed systems.
- Existing deployed systems and updates.
- Migrated systems and associated changes.

The Contractor shall create comprehensive training plans, manuals, documentation, and training aids. Electronic training tools such as video teleconferencing and computer-based training shall be utilized to enhance the effectiveness of training materials and courses. Training shall properly operate, maintain, and test systems, applications, and products. Additionally, The Contractor shall provide knowledge transfer to technicians and staff regarding services and products delivered under any functional areas described herein. Training shall enable personnel to operate and maintain the product or processes independently in the future. The Contractor shall identify and/or provide any additional training required by end-users, technicians, or any other staff for implementation, maintenance and use of deliverables specified in individual TOs.

##### C.4.21.1 Cybersecurity Training and Curriculum Design

Cybersecurity training plans and human-centered curriculum design shall align with NIST SP 800-53 controls AT-2 (Awareness Training) and AT-3 (Role-Based Training). The Contractor shall incorporate human-factor and cyber psychology principles addressing cognitive bias, decision-making under stress, fatigue, and risk perception into NICE-aligned role curricula. Training content shall reinforce secure behaviors, sound judgment under pressure, and accountability for users, privileged roles, and executives. Sustainment activities shall ensure curricula remain current based on threat trends, workforce performance indicators, and control assessment results.

#### **C.4.21.2 Scenario-Based Exercises**

Discussion-based, scenario-driven tabletop exercise shall be designed and conducted in alignment with PM-14 (Testing, Training, and Monitoring) and IR-2 (Incident Response Training). The Contractor shall develop scenarios that test both technical response and human decision dynamics, including but not limited to:

- Communication breakdowns.
- Authority ambiguity.
- Insider risk indicators.
- Escalation behaviors.

Exercises shall evaluate participant performance against defined roles, responsibilities, and behavioral expectations. Sustainment shall incorporate exercise outcomes into revised training, procedures, and operational controls.

#### **C.4.21.3 Cybersecurity Workforce Certification Management**

Certification training planning and management services shall align with AT-4 (Training Records) and PM-12 (Insider Threat Program) to ensure traceability, accountability, and behavioral risk awareness. The Contractor shall:

- Plan, manage, and track certification and continuing education activities mapped to workforce roles and access levels.
- Monitor for skill gaps, anomalous behavior trends, and refresher triggers based on incidents or control weaknesses.

Sustainment shall ensure certifications and training pathways remain current with evolving control requirements and workforce risk profiles.

#### **C.4.21.4 Adaptive Behavior User Training**

Adaptive user behavior enforcement and corrective training services shall be designed, developed, deployed, and sustained to detect, prevent and remediate attempted violations of cybersecurity standards by users and privileged users in real time. The Contractor shall:

- Implement controls that automatically block non-compliant actions.
- Present contextual guidance and require completion of targeted, role-appropriate training before access or activity is restored.
- Integrate these capabilities with enterprise identity, access, and monitoring systems for consistent enforcement across all environments.

The system shall automatically record training failures, policy violations, user actions and remediation actions within enterprise training and risk management repositories. Sustainment



shall ensure continuous refinement of enforcement logic, training content, and reporting based on observed behavior and evolving cybersecurity requirements.

#### **C.4.22 Information Technology Facilities**

The Contractor shall provide a comprehensive IT solution to the agency, which includes incidental facility design and modification services, site surveys, facility connectivity, and installation. Specific tasks, requirements, deliverables and performance expectations will be defined in individual Task Orders (TOs).

##### **C.4.22.1 Incidental Facility Design and Modification Services**

The Contractor shall deliver infrastructure design, installation, and modification services necessary to support the IT solution. These activities may include, but are not limited to:

- Modifying rooms or buildings at existing sites (e.g. field, region, HQ, collocation data centers, and other designated sites and offices) to support the information transport infrastructure.
- Furnishing and installing Unshielded Twisted Pair (UTP) (e.g. Cat-6, 6A, 8, and newer) Cabling.
- Single-Mode (SM), Multimode (MM), and newer Fiber Optic (FO) cabling.
- Providing telecommunications pathways and spaces, work area outlet terminations, patch panels, racks, and cabinets.
- Implementing fire-stop, fire suppression, telecommunications grounding and bonding.
- Designing and installing fire-suppression systems.

Affected rooms or buildings will host IT systems and provide work areas for operating personnel. Activities may also include dismantling and removing existing infrastructure to enable modifications. All work shall be coordinated with the appropriate organization prior to the issuance of the Service Order (SO) and executed withing approved parameters. Reviews may include compliance with master plans, land use plans, utility systems capacity, and environmental constraints.

Associated activities will be limited to incidental facility modifications related to the project and involve minimal real property maintenance, repair, and modification activities. All efforts shall comply with Section 2.0 - Applicable Documents, and all federal and state standards.

##### **C.4.22.2 Site Surveys**

The Contractor shall perform the site surveys to develop comprehensive plans for installing information transport systems and IT work areas. Surveys shall provide input to Fixed Station Configuration Management Plans (CMP), including equipment reconfiguration requirements. Deliverables shall include but limited to:

- Fully developed and dimensioned floor plan layouts.



- Bills-of-material.
- Telecommunications pathways and spaces.
- COOP/DR considerations.
- Cabling, power distribution, environmental conditioning.
- Test and cutover plans.
- Grounding, access floor systems, lighting, backboards.
- Labor estimates.
- Required Government Furnished Equipment (GFE) and materials.

#### **C.4.22.3 Facility Connectivity**

The Contractor shall design and install IT distribution systems, which may include:

- Approved inside plant fiber and copper media.
- Media connectors, patch panels, fiber distribution cabinets.
- Patch cords, pre-terminated cable assemblies.
- Entrance facilities, backbone (first and second level), horizontal distribution.
- Termination blocks, wireless network components, cross-connects, and inter-connects.

Knowledge of outside plant and aerial distribution methods may be required.

#### **C.4.22.4 Installation**

The Contractor shall install hardware and software/firmware as specified in individual TOs. Installation may involve:

- Fabrication of mounts, brackets, and installation kits.
- Cabling, connections, and interconnecting devices.

The Contractor shall assist the Government in identifying all equipment and utilities required for installation, including GFE/GFM. The Government, with Contractor assistance, shall ensure that the required equipment, utilities, and resources are available at the installation site.

#### **C.4.22.5 Physical Security Systems**

The Contractor shall develop, implement and/or maintain the physical security functions to including but not limited to:

- Building access guides.
- Restricted access levels to facilities.
- Biometrics.
- Alarm systems.

## **C.5 DELIVERABLES**

### **C.5.1 Products**

All products shall be delivered to the designated Government locations and accepted by authorized Government personnel as specified in the contract and individually awarded TOs. Inspection and acceptance criteria shall be clearly defined within each individually awarded TOs. The Contractor shall promptly notify the COR and IT P/PM of any discrepancies identified during acceptance inspection.

### **C.5.2 Data**

The Government shall receive Unlimited Rights to all intellectual property first produced and delivered under this contract and individually awarded TOs, in accordance with FAR 52.227-14, Rights In Data - General (MAY 2014). If specified at the contract and individually awarded TOs level, this may include full ownership rights to source code and all supporting documentation.

License rights for any Commercial Computer Software shall be governed by FAR 52.227-19, Commercial Computer Software License (DEC 2007). All data delivered shall be submitted and protected in accordance with HUD Handbook 3500.1 - HUD Source Code Policy.

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## **C.6 SECURITY AND PRIVACY**

### **C.6.1 Information Security and Privacy Security Requirements**

The Contractor shall comply with HUD's security requirements.

### **C.6.2 Personnel Security Requirements**

The Contractor shall comply with all HUD personnel security requirements specified in this contract and in each individually awarded TOs. All Contractor personnel requiring access to HUD sensitive information or computer systems shall undergo background investigations and must receive a favorable determination from HUD prior to being granted access. The Contractor shall:

- Prescreen Personnel – Ensure all personnel requiring system access maintain the appropriate background investigation status before assignment.
- Submit Personnel Roster - Provide a roster of contractor and subcontractor personnel within three (3) business days after contract or TOs award to the COR to initiate background investigations. Update the roster within one (1) business day of any staffing changes and submit updates to the COR.
- Provide Required Information - Submit all requested information related to employee background investigations within three (3) business days of a request from the COR and/or HUD's PSD staff.
- Assume Responsibility for Personnel Actions – The Contractor shall be fully responsible for the actions of all contractor and subcontractor personnel under this contract and individually awarded TOs. In the event damages arising from Contractor personnel actions, the Contractor shall provide all necessary resources (e.g. funding, material, etc.) to remedy the incident.
- Withdraw Personnel When Required – Immediately withdraw any employee from consideration upon notification by the COR and/or HUD PSD of an unfavorable adjudicated background investigation.
- Ensure Compliance – Adhere to all personnel security investigation requirements. Failure to comply may result in loss of physical and logical access to HUD facilities and system, and/or termination of the contract or individually awarded TOs for default.
- Manage PIV Card Compliance - Ensure all PIV card holders follow HSPD-12 policies and HUD procedures including:
  - Proper use and protection of identity credentials.
  - Displaying badges at all times while working in HUD offices, collocation data centers, and other designated facilities.
  - Returning PIV cards upon termination of employment or contract relationship.
- Comply with Additional Security Requirements – Follow all additional security requirements specified in each of the individually awarded TOs.

### **C.6.3 PIV Cards**

Contractor employees working at any HUD facility shall display a Personal Identity Verification (PIV) card on their person at all times. It is the Contractor's responsibility to request and obtain PIV cards from the COR prior to the employee's first workday. The Contractor shall:

- Return PIV Cards – All PIV cards must be returned to the COR on the same day an individual's employment ends and upon termination of the contract or any individual TOs.
- Report Lost Cards Immediately – The Contractor shall notify the COR immediately of any lost or stolen PIV cards.

### **C.6.4 Classified Work**

Work performed under this contract or individual TOs may involve secure networks, facilities, and sensitive information. Specific security requirements and suitability determination will be defined in each individual TOs.

The Contractor should anticipate the need to provide personnel with security clearances up to Top Secret, as required by the TOs. Contractors must possess the appropriate clearance at the TOs level for proposal purposes.

### **C.6.5 Incident Reporting and Management**

The Contractor shall notify the COR, IT P/PM, and the assigned Information Security Officer (ISO) of any security events, and the Privacy Officer (PO) for any privacy violations, with one (1) hour of occurrence. The Contractor shall provide continuous updates on all reported security and privacy events until the issue is formally closed by the ISO or PO.

### **C.6.6 Security and Privacy Awareness Training**

The Contractor shall complete initial security and privacy awareness training and acknowledge acceptance of the HUD Rules of Behavior (ROB) within two (2) business days of receiving the contract or TOs awards. Additionally, the Contractor shall complete annual security and privacy awareness training and reaffirm acceptance of the HUD ROB prior to its expiration.

### **C.6.7 Security Training**

The Contractor shall complete all assigned security role-based training within three (3) business days of assignment by the COR. Completion of this training is a prerequisite for receiving elevated system privileges.

## C.7 REPORTING AND MEETING REQUIREMENTS

### C.7.1 Reporting Requirements

The deliverables defined below are required to be compliant with this contract and/or individual TOs as specified and shall be submitted electronically to the H2NG Microsoft Team site or to the individual TOs specific Microsoft Teams site. **Email submissions will not be accepted.**

The Contractor shall submit one quarterly Small Business Participation Report and one quarterly Veterans Employment Certification Report which includes a rollup of the data from each TOs. These reports are due on January 15th (for the period October through December); April 15th (for the period January through March); July 15th (for the period April through June); and October 15th (for the period July through September). The Contractor shall submit a Contractor's Progress, Status, and Management Report; Contract Performance Report; Status of Government Furnished Equipment Report; and Personnel Contractor Manpower Report for each individual TOs on a quarterly basis, unless otherwise specified by the individual TOs's PWS. These reports shall be submitted three (3) months after the start date of each individual TOs and quarterly thereafter. If the Government rejects a quarterly report, the Contractor shall have five (5) business days to remediate any Government comments and resubmit the report in the TOs Microsoft Teams site. Deliverables with due dates falling on a weekend or holiday shall be submitted the following Government business days after the weekend or holiday.

#### C.7.1.1 Contractor's Progress, Status and Management Report

The Contractor shall submit a quarterly Contractor's Progress, Status, and Management Report for each TOs awarded. This TOs report shall be unique to each TOs. These reports shall convey the status of the TOs performance. All relevant billing/invoice information shall be posted to the IPP portal. TOs that are completed shall be listed as such. A standard format is set forth, and shall be utilized for submission of the below required information.

For each TOs, the Contractor shall indicate/discuss:

- TOs summary.
- Performance metrics.
- TOs schedule.
- VIP / PLM Compliancy (as applicable).
- Critical items for Government review.
- Accomplishments.
- An itemized listing of all Information and Communication Technology (ICT) deliverables and their current Section 508 conformance status.
- Significant open issues, risk and mitigation action.
- Summary of issues closed.
- Meetings completed.
- Projected meetings.
- Subcontractor performance – discuss 1<sup>st</sup> tier Subcontractor(s) performance.

- Projected activities for next reporting period.
- Explanation if the reporting period is over three months.
- Invoice/receiving reports submitted.
- Milestone payment schedule, if applicable.
- Set-aside expenditures, if applicable.

#### **C.7.1.2 Contract Performance Report (CPR)**

The Contractor shall submit a quarterly Contract Performance Report for each Time and Materials or Cost Reimbursable type TOs awarded. This TOs report shall be unique to each TOs. This report is not required for Firm Fixed Price TOs. Contractors may be required to support EVMS (Earned Value Management System) at the TOs level. The Contract Performance Report, as set forth in Section J, Attachment 004 (Time and Materials) and Attachment 005 (Cost), shall be submitted quarterly via the TOs Microsoft Teams site.

For Each Time and Materials Task, the Contractor shall indicate:

- Expenditures for the reporting period by labor, material and ODCs.
- Labor costs shall be broken down by assigned numbering system for contract, TOs and labor category, entity (Prime or Subcontractor), rates and hours.
- Material costs and ODCs shall be identified by type, and subcontractor (as applicable), and discussed.
- Total task expenditures for the fiscal year to date, indicated as total, labor, materials and ODCs.
- Total task expenditures since task award, indicated as total, labor, materials and ODCs.

For Each Cost Reimbursable Task, the Contractor shall indicate:

- Labor costs broken down by assigned numbering system for contract, TOs and labor category, skill level, entity (Prime or Subcontractor), rate and hours, material costs, ODCs, Cost of Money and fee.
- Total task expenditures for the fiscal year to date, indicated as total labor, materials, ODCs, Cost of Money, and fee.
- Total task expenditures since task award, indicated as total, labor, materials, ODCs, Cost of Money, and fee.

#### **C.7.1.3 Status of Government Furnished Equipment (GFE) Report**

The Contractor shall submit a quarterly Status of GFE Report for each TOs awarded. This TOs report shall be unique to each TOs. The GFE Report, as set forth in Section J, Attachment 006, shall be submitted quarterly via the TOs Microsoft Teams site.

- A. TOs.
- B. Project Name.
- C. Type of Equipment.

- D. Tracking Number.
- E. HUD Bar Code.
- F. Location.
- G. Value.
- H. Total Number of Pieces.
- I. Total Value of Equipment.
- J. Anticipated Transfer Date to Government.
- K. Anticipated Transfer Location.

#### **C.7.1.4 Personnel Contractor Manpower Report**

The Contractor shall submit a Personnel Contractor Manpower Report (MS Excel) for each TOs awarded, on a quarterly basis listing all personnel for each TOs awarded. The reports shall be unique to each TOs. The Personnel Contractor Manpower Report, as set forth in Section J, Attachment 008, shall be submitted quarterly via the TOs Microsoft Teams site. The information required is as follows:

- A. TOs.
- B. Employee Name.
- C. Background Investigation/Clearance level and/or Status.
- D. Company name.
- E. Prime/Subcontractor.
- F. Labor Category.
- G. Facility location.
- H. Tour of Duty Schedules (e.g. Monday through Friday, 9:00 am to 5:00 pm).
- I. Contractor Rules of Behavior.
- J. HUD Privacy and Information Security Awareness and Rules of Behavior Training.
- K. Annual HUD Privacy Training.
- L. Universal Unique Identifier UUID (Badge Number bottom right of back of badge) .
- M. Facility Issuing Badge.
- N. Badge Expiration Date.
- O. Date Disassociated From TOs (for employees who no longer support this TOs) .
- P. Date Badge Returned to COR; and/or:
  - i. For employees transitioned to new/follow-on TOs, the TOs number the badge is associated with.

#### **C.7.1.5 Contractor Staff Roster**

The Contractor shall submit a Contractor Staff Roster for each TOs awarded, of Contractor and Subcontractor employees within three (3) business days after each TOs award for all personnel employed under each TOs to begin their background investigations. As personnel changes occur a revised roster is required. The Contractor Staff Roster shall be updated and delivered only to the COR within one (1) day of any changes in employee status, training certification completion status, Background Investigation level status, additions/removal of employees, etc., throughout the Period of Performance. Do not post the Contractor Staff Roster to the TOs Microsoft Teams site. The Contractor Staff Roster should indicate which employees are active or inactive. For

inactive employees, the roster should indicate the date the employee was separated from the TOs and their credentials returned to the COR.

The Contractor Staff Roster shall contain:

- A. Contractor's Full Name.
- B. Email Address.
- C. Place of Birth.
- D. Date of Birth.
- E. Security/Privacy Training Completion Dates.
- F. Risk Designation-individual background investigation level requirement (Refer to Section 4.6 of the TOs PWS for investigative requirements by task).
- G. Existing Background Investigation and/or Clearance (if applicable).
- H. The Contractor shall submit full Social Security Numbers either within the Contractor Staff Roster or under separate cover to the COR. The preferred method to send the Contractor Staff Roster or Social Security Number is by encrypted e-mail. If unable to send encrypted e-mail, other methods which comply with FIPS 140-2 are to encrypt the file, use a secure fax, or use a traceable mail service.

#### **C.7.1.6 Small Business Participation Report**

The Contractor shall submit the Small Business Participation Report, on a quarterly basis.

The Small Business Participation Report shall contain:

- a. Company Name.
- b. Company Cage Code/EUI Number.
- c. Company Size.
- d. H2TNG Prime Contract Number.
- e. Date Report Submitted.
- f. POC for questions on this report (name, phone, email).
- g. Small Business Participation Quarter and Year.
- h. Total Obligated Dollars (Cumulative through the end of the reporting quarter. e.g. 3/31, 6/30, 9/30, 12/31).
- i. Funded/Obligated Large Business Dollars.
- j. Funded/Obligated Small Business Dollars and Total Obligated Dollars for Total Small Business Subcontracting Participation (Goal 1 Small Business Actuals).
- k. Small Business Category Cumulative Funded/Obligated Dollars and Total Obligated Dollars for (GOAL 2 - Breakout of Small Business Dollars on Total Contract Obligated Dollars):
  - i. Small Disadvantaged Business.
  - ii. Women Owned Small Business.
  - iii. HUBZone.
  - iv. Veteran Owned Small Business.
  - v. Service-Disabled Veteran Owned Small.
  - vi. Small Business (No other category above).



- l. Location where data and records for this report are housed and maintained.
- m. Explanation (as necessary). Provide detailed explanations for percentages less than what was proposed. Also explain dollars not obligated to either a large or small business (i.e., unallocated obligations).
- n. Subcontractor Identification:
  - i. Subcontractor Name.
  - ii. Subcontractor Status.
  - iii. Total Dollars Obligated (Cumulative).
- o. Verification of submission of Subcontract awards in the Federal Funding Accountability and Transparency Act Subaward Reporting System (FSRS).

#### **C.7.1.7 Veterans Employment Certification Report**

The Contractor shall provide a Veterans Employment Certification Report, on a quarterly basis via the ATOMS portal listing the following:

- A. Total number of employees at time of proposal submission.
- B. Total number of Veterans employed at time of proposal submission.
- C. Total number of current employees.
- D. Total number of current Veterans employed.
- E. Location where data and records for this report are housed and maintained.
- F. The Contractor's Chief Executive Officer (CEO), or, for a joint venture, the CEO for both parties that make up the joint venture, or equivalent, shall certify in the Veterans Employment Certification Report that the Contractor has, in good faith, relied on the representations of its employees to derive the total number of employees and the number of Veterans that it employs. This certification shall be as of the last day of the quarter being reported.

#### **C.7.2 Meetings and Reviews**

For successful management and contract surveillance, the following meetings and reviews are required.

##### **C.7.2.1 Project Office Initial Program Review (IPR)**

The Government shall host an IPR within 30 days after this TOs and/or individual TOs awards to review the PWS, business policies, and procedures, and introduce personnel.

##### **C.7.2.2 Post-Award Conferences**

The Government intends to convene a Post-Award Conference with each awardee within 60 days after this task and/or individual TOs award(s). The CO shall notify all Prime Contractors of a specific date, location and agenda within thirty (30) days after contract award.

#### **C.7.2.3 Program Reviews**

At the discretion of the CO, Program Management Review (PMR) Meetings shall be conducted by the Contract Specialist and/or designated COR for each contract. Dates, locations, agenda, and attendance requirements shall be specified by the appropriate Government representative, at least five (5) calendar days prior to the meeting.

#### **C.7.2.4 Quarterly Collective Prime Program Reviews**

The Government shall host a quarterly Prime Program Management Review (QPPMR) Meeting with the designated Prime Program Manager and one additional Contractor attendee. Dates, locations, and agenda shall be specified at least five (5) calendar days prior to the meeting.

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